

Problems in Graph Sampling and Graph Learning

A Thesis

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by

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Abstract

We address the challenge of learning a Graphical Representation for the Entities (also called Nodes) present in a population, each characterized by its own Multivariate Features (also called Signals). Traditionally in Graph Signal Processing (GSP), to tackle this problem, smoothness of the signals in the graph spectral domain was a ubiquitous assumption, signifying that the Signals vary Smoothly over the Nodes, but this assumption would only have been true if the signals observed over the nodes resulted from a Pure Diffusion Process. While at the onset of research in this field of GSP, an assumption like this might have sounded reasonable, but to be more pragmatic to real-world use cases, and as an advancement to this field of GSP, we propose a Novel Formulation for Graph Learning, that encourages Sparsity in the Graph Spectral Domain. Further, we rigorously test our approach on synthetic and real-world datasets, demonstrating its effectiveness. Additionally, due to inconsistency in thresholding of the learned Matrices as a post-processing step taken by other graph learning algorithms, we propose a principled approach for hyperparameter selection which is applicable not only to our Formulation but also to other GSP-based Graph Learning Formulations in general.

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गुरुर्ब्रह्मा गुरुर्विष्णुः गुरुर्देवो महेश्वरः।

गुरुः साक्षात् परं ब्रह्म तस्मै श्री गुरवे नमः।।

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Chapter 1

Introduction

1.1 Subset Selection Problem

Selection of the *right* subset, traditionally enjoys the legacy of being one of the most sought-after problem. Thanks to the ever-expanding world, which makes this problem never lose its charm. Choosing the best representative from a large domain is a problem that fundamentally being ubiquitous and NP-Hard at the same time, suggests always in all contexts where this problem arises, that there is a pressing need for optimizing the procedure of selecting the right subset. The problem has seen many applications to it occurring almost every possible walks of life, be it from picking up the right subset of points to pick from the set of points, to appoint as the cluster heads, and then dispersing off the rest of the points to those zonal heads, which could be commonly called as clustering or picking up the right representative subset from the population which will help identify the full population dynamics. All in all this problem of Subset selection emphasizes strongly an optimization problem, which wants to maximize a function from the minimum possible observatory or scientific effort.

CHOOSING THE CLUSTERING “HEADS” TO BE THE REPRESENTATIVES

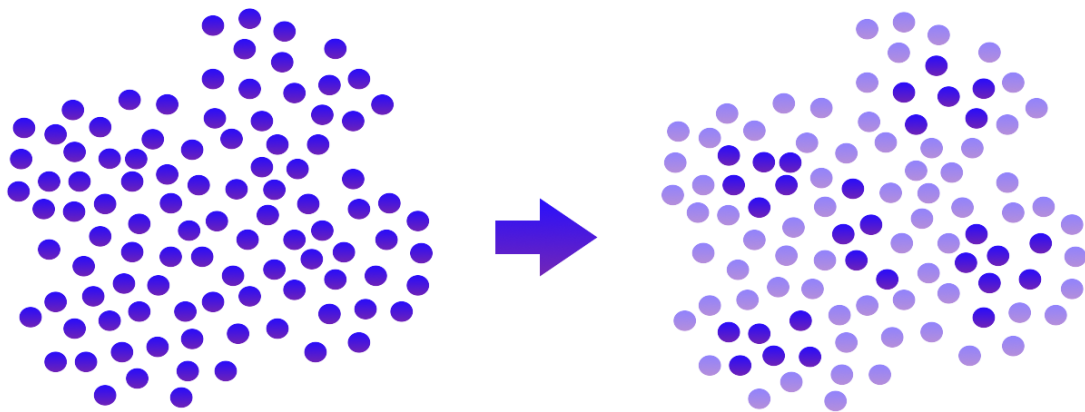


Figure 1.1: Picking Up the “Right” Subset

Presented here is yet another application of Subset selection in the context of Speedy group testing on the verge of an Epidemic. An interesting primer on epidemics and epidemiology is presented below: An epidemic is like a sudden storm of sickness that sweeps through a community, leaving a trail of coughs, fevers, and aches in its wake. It’s like a wildfire of infection, spreading rapidly and threatening to overwhelm the healthcare system. It’s a wake-up call that reminds us

of the power of infectious diseases and the importance of public health measures to keep them at bay. Epidemiology is like a detective story, but instead of solving crimes, the goal is to unravel the mysteries of diseases and outbreaks. It's a field of study that combines the power of science and data to understand how diseases spread, who is most vulnerable, and how to prevent and control them. Epidemiologists are like disease detectives, using their skills to track down clues and patterns, piece together evidence, and make sense of complex health issues. With epidemiology on our side, we can uncover the secrets of diseases and protect our communities from their deadly grip.

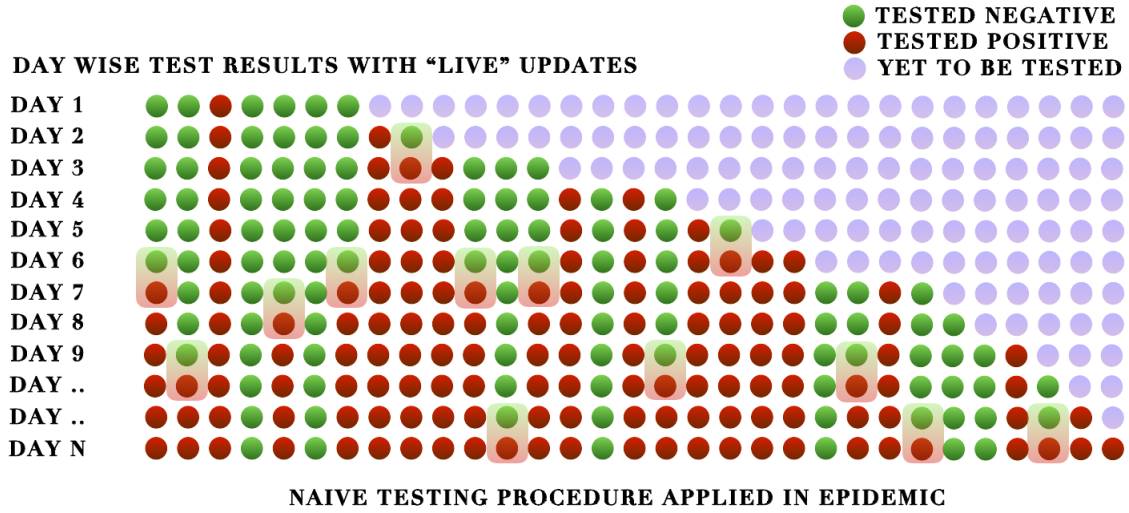


Figure 1.2: The *Testing* “PARADOX”

1.2 Optimal Case Load Estimation Problem

In the entire Epidemiological narrative, it's illogical to proceed about almost anything unless we know exactly who has caught up the disease and who hasn't. So it's quite succinct that one of the most vital element to begin any sort of epidemiological modeling is “*Testing*”. Obviously, the naive approach is to begin testing every single individual in the region and then tinker with ways to curb the viral outbreak. But the irony is that by the time we reach to test the 800th person we might find the 1st person to whom when we tested earlier in the beginning and tested negative, has been caught with the disease. So all efforts got wasted? Should we restart the entire testing campaign? or Should we just not care about all this and take action based on whatever data is collected? The need of the hour is not to answer these questions but to thoroughly understand those questions, to understand exactly where all this is going, and understand the subtleties lying underneath the core problem. All the above was just regarding one such typical problem that might be encountered while going about Testing, there could be tons of other problems waiting to be questioned and solved. A principled approach to testing is paramount and should be regarded as the highest priority task at hand, in the World of epidemiology.

1.3 Introduction to Graph Signal Processing

In this Chapter we will talk in general about Graph Signal Processing and Graph Laplacian.

1.3.1 Graph Signal Processing

Graph signal processing (GSP) is a field that deals with signals defined on graphs, where graphs are mathematical objects consisting of vertices and edges that model relationships between different entities. In GSP, signals are represented by vectors whose components correspond to values assigned to vertices in a graph.

One of the key challenges in GSP is to develop methods that can handle the high-dimensional nature of signals on graphs. Since graphs can have a large number of vertices and edges, signal processing techniques that are designed for low-dimensional data may not be directly applicable to graphs.

To overcome this challenge, GSP researchers have developed a variety of techniques that leverage the underlying graph structure. For example, the graph Laplacian matrix, which encodes information about the relationships between vertices, has been used to define graph Fourier transforms and associated frequency-domain representations of graph signals. This has allowed researchers to apply tools from classical Fourier analysis to the graph setting, enabling them to develop new filtering, sampling, and compression methods for graph signals.

Another key area of research in GSP is the development of graph learning algorithms, which aim to learn the underlying graph structure from data. This is an important task in many applications, such as social network analysis, where the graph structure may not be known a priori.

Overall, GSP is an exciting and rapidly growing field that has the potential to revolutionize signal processing by providing powerful tools for analyzing and processing signals on graphs.

1.3.2 Graph Spectral Domain

Spectral graph theory has traditionally focused on the construction, analysis, and manipulation of graphs, rather than signals on graphs. It has proven to be particularly useful in the creation of expander graphs, graph visualization, spectral clustering, graph coloring, and many other applications in chemistry, physics, and computer science.

In the field of signal processing on graphs, spectral graph theory has been utilized as a tool to define frequency spectra and expansion bases for graph Fourier transforms. In this section, we will provide a brief overview of some fundamental definitions and notations from spectral graph theory, emphasizing how it enables us to extend many of the important mathematical concepts and intuitions from classical Fourier analysis to the graph setting.

1.3.3 Graph Laplacian

The unnormalized graph Laplacian, also known as the combinatorial graph Laplacian, is defined as:

$$L = D - W \tag{1.1}$$

where D is a diagonal matrix with the i th diagonal element d_i equal to the sum of the weights of all the edges incident to vertex i , and W is the adjacency matrix of the graph.

The Laplacian is a difference operator for any signal f in $\mathbb{R}^N$, and it satisfies:

$$(Lf)(i) = \sum_{j \in N_i} (W_{i,j}[f(i) - f(j)]) \tag{1.2}$$

where N_i is the set of vertices connected to vertex i by an edge, and $N(i, k)$ is the set of vertices connected to vertex i by a path of k or fewer edges.

The graph Laplacian L is a real symmetric matrix with a complete set of orthonormal eigenvectors denoted as $u_0, u_1, \dots, u_{N-1}$. The associated eigenvalues are real and nonnegative, denoted as $\lambda_0, \lambda_1, \dots, \lambda_{N-1}$, satisfying $Lu_i = \lambda_i u_i$ for $i = 0, 1, \dots, (N-1)$. The eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The number of connected components of the graph determines the multiplicity of the eigenvalue 0, and since we consider connected graphs, we assume that the graph Laplacian eigenvalues are ordered as $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{N-1}$. The set of all eigenvalues is denoted as $v(L) = \lambda_0, \lambda_1, \dots, \lambda_{N-1}$.

1.4 Compressed Sensing: Background and Motivation

Compressed sensing (CS) is a powerful framework for reconstructing unknown signals from sparse measurements. It has been widely used in various signal processing applications, including image processing, signal acquisition, and data compression. In the context of graph signal processing, CS has been applied to reconstruct graph signals from a limited number of measurements.

Compressed Sensing Theory for Reconstruction of Partially Known Signals:

The basic idea of CS is to leverage sparsity, or compressibility, of signals in some basis or transform domain. The key observation is that many real-world signals can be represented by a sparse or compressible linear combination of basis functions. In other words, most of the signal energy is concentrated in a few basis coefficients, while the rest of the coefficients are close to zero. CS exploits this sparsity property by acquiring a small number of linear measurements of the signal and then using an appropriate reconstruction algorithm to recover the original signal from the measurements. In the graph signal processing setting, the graph Laplacian matrix plays a central role in CS-based signal reconstruction. The graph Laplacian encodes the structural information about the graph, including the connectivity and the degree of each vertex. It also provides a natural framework for defining a graph Fourier transform and associated frequency-domain representations of graph signals.

To apply CS to graph signals, one can represent the graph signal in the graph Fourier domain, where the signal is sparse or compressible. The sparse representation can be obtained by solving an optimization problem that minimizes the l1-norm of the graph signal in the graph Fourier domain subject to the constraint that the measurements are consistent with the acquired data. The resulting optimization problem is a convex optimization problem that can be efficiently solved using various algorithms such as iterative soft thresholding (IST) and orthogonal matching pursuit (OMP). The key challenge in applying CS to graph signals is to design appropriate measurement matrices that allow for accurate signal reconstruction from a small number of measurements. In general, the measurement matrix should be incoherent with the graph Fourier basis and should satisfy certain properties such as the restricted isometry property (RIP) to ensure accurate signal recovery.

In summary, compressed sensing theory has been applied to graph signal processing by leveraging the sparsity property of graph signals in the graph Fourier domain. The graph Laplacian matrix provides a natural framework for defining the graph Fourier transform and associated frequency-domain representations of graph signals. CS-based signal reconstruction can be achieved by solving a convex optimization problem that minimizes the l1-norm of the graph signal in the graph Fourier domain subject to the constraint that the measurements are consistent with the acquired data. Appropriate measurement matrices that are incoherent with the graph Fourier basis and satisfy certain properties such as RIP are essential for accurate signal recovery.

Chapter 2

Optimal Case Load Estimation Problem

The aim of this research is to estimate the caseload within a given population, which could encompass either the viral load carried by each individual or the number of cases reported from specific locations such as hospitals or villages. This estimation process will incorporate additional demographic data, including geographical networks and mobility information. By leveraging this approach, the goal is to provide a more accurate prediction of caseloads.

To illustrate, consider the task of predicting the number of COVID-19 cases that might arise in a large urban area such as Mumbai. By selecting a representative subset of approximately six to seven regions within Mumbai, we can gather detailed case load data from these areas. This subset will serve as the foundation for extrapolating predictions for the entire region.

The methodology involves analyzing the caseload data from the selected regions and integrating it with relevant demographic information. This could include contact tracing data, transportation patterns, and other mobility-related information. By doing so, we can create a comprehensive model that reflects the dynamics of COVID-19 spread within these representative regions.

Ultimately, this model aims to provide accurate estimates of potential caseloads in the remaining areas of Mumbai. The integration of detailed caseload data with demographic and mobility information is expected to enhance the precision of these predictions, thereby contributing to more effective public health planning and response strategies.

2.1 Compressed Sensing Model for Optimal Reconstruction

We will build this model as illustrated in the aforementioned example. Let $X \in \mathbb{R}^M$ where M could represent the total number of regions taken into consideration in Mumbai as per the example and $Y \subseteq X$, representing a few of those M regions.

$$\text{size}(Y) = K$$

Let C be a Subset Selector Matrix or in more technical terms a ‘Sensing Matrix’. Based on the subset Y , we want to reconstruct X . So the Corresponding formulation should look something like this: Optimize for X :

$$\|Y - C \cdot X\|_2^2$$

But this has infinitely many solutions. So how could we obtain X uniquely? How can CS help in this? CS takes advantage of the fact that many signals are sparse, i.e., they contain many coefficients close to or equal to zero, when represented in some domain.

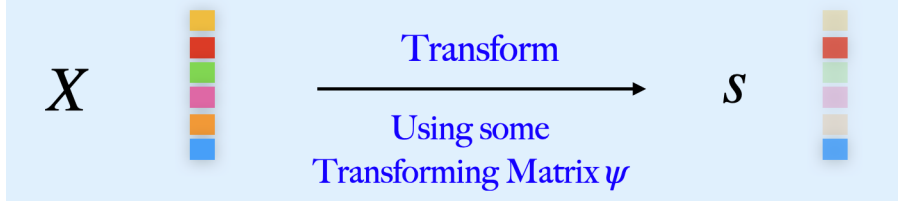


Figure 2.1: Spectral Representation of the Signals

$$X \cdot \psi = s$$

Law of Conservation of Energy $\implies$ Total Energy in this system should be preserved!

$$X = \psi^{-1} \cdot s$$

$\implies$ Components of ψ^{-1} must capture the notion of energy/frequency.

Now, where can we get this notion of energy/frequency in ψ ? Based on the visual representa-

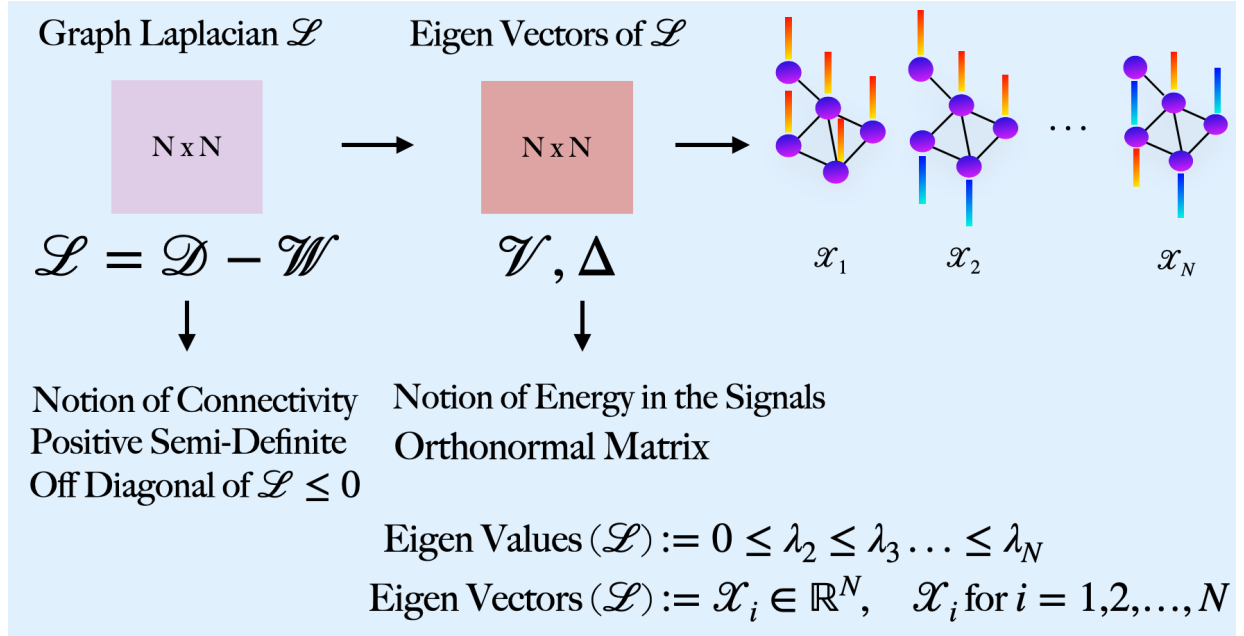


Figure 2.2: Notion of Frequency

tion of the Components of the eigenvectors starting from lower to higher eigenvalues, it's succinct that the notion of change is indeed being captured by them! hence the following will thus follow naturally:

$$\begin{aligned} \psi^{-1} = \mathcal{V} &\implies X = \mathcal{V} \cdot s \\ Y = C \cdot X &\implies Y = C \cdot \mathcal{V} \cdot s \end{aligned}$$

CS says not all under-determined systems of linear equations have a sparse solution. However, if there exists a unique sparse solution to the under-determined system, then the compressed sensing framework allows the recovery of that solution. So, $X = \mathcal{V} \cdot s$, i.e., exact recovery is only possible when s is the sparsest. Thus, for a nice recovery, we propose the following:

$$s^* = \min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

And the 'Recovery' is given by:

$$X^* = \mathcal{V} \cdot s^*$$

2.2 Measles Outbreak in Mumbai 2022

2.2.1 What is Measles?

A Disease which Spreads easily from Person to Person through Air, and can cause Fever, Cough, Rash.

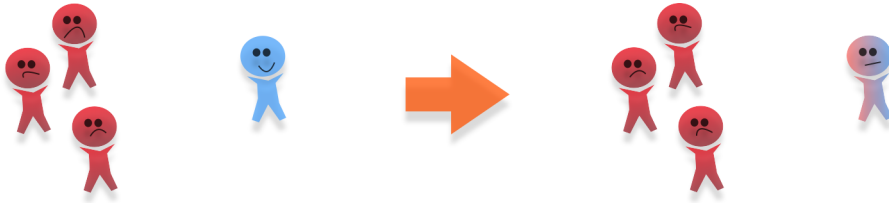


Figure 2.3: Measles Disease

2.2.2 Analysis of the Dataset from BMC

The work presented herein focuses on ‘Testing’ within the context of Measles disease spread. This research utilizes a dataset provided by the Brihanmumbai Municipal Corporation (BMC), which serves as a comprehensive source of information for analyzing the spread of Measles. In the following sections, we provide a detailed breakdown of all potentially relevant information that can be derived from this dataset. By examining various aspects such as infection rates, demographic factors, and geographical distribution, this analysis aims to uncover patterns and insights that could inform public health strategies and intervention measures. The dataset’s richness allows for an in-depth exploration of Measles transmission dynamics within Mumbai, contributing valuable knowledge to the field of epidemiology. The findings from this study can aid in the development of targeted approaches to mitigate the spread of Measles and improve overall public health outcomes.

Age Bin	Average	Min	Max	Count
0-1	0.63	0	0.92	349
1-2	1.38	1	1.92	238
2-3	2.37	2	2.92	131
3-4	3.42	3	3.92	88
4-5	4.43	4	4.92	90
5-6	5.33	5	5.92	64
6-7	6.42	6	6.92	53
7-8	7.35	7	7.92	62
8-9	8.4	8	8.83	37
9-10	9.41	9	9.92	33
10-12	10.81	10	11.92	51
12-16	13.47	12	15.67	30
16-20	17.93	16.08	19.92	20
20-25	21.93	20	24.75	18
25-30	28.17	25.67	29.92	8
30-40	32.125	30.25	35.75	6
40-50	42.73	40.42	47.33	5
50-60	50.63	50	51.25	2
60-70	62.5	60	65	2

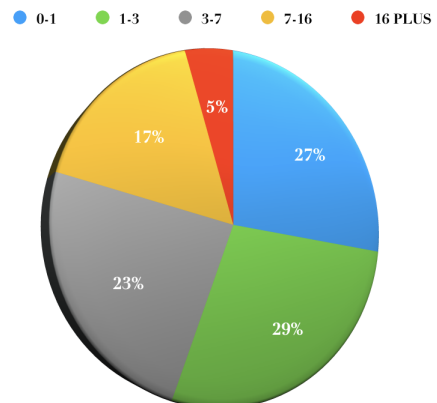


Figure 2.4: Age Wise Statistics

The pie chart, accompanied by the summarized dataset, presents an age-wise breakdown of the information derived from the BMC-provided dataset. This visual representation clearly indicates a significant dominance of cases among infants compared to adults. The chart highlights the age distribution of individuals affected by Measles, revealing a disproportionately higher number of infections in the infant population. This finding underscores the critical importance of focusing public health interventions and vaccination efforts on protecting the youngest and most vulnerable members of the community. The detailed age-wise analysis provided by the dataset allows for a more targeted approach in addressing the spread of Measles, ensuring that resources are allocated effectively to mitigate the impact on infants and reduce overall infection rates.

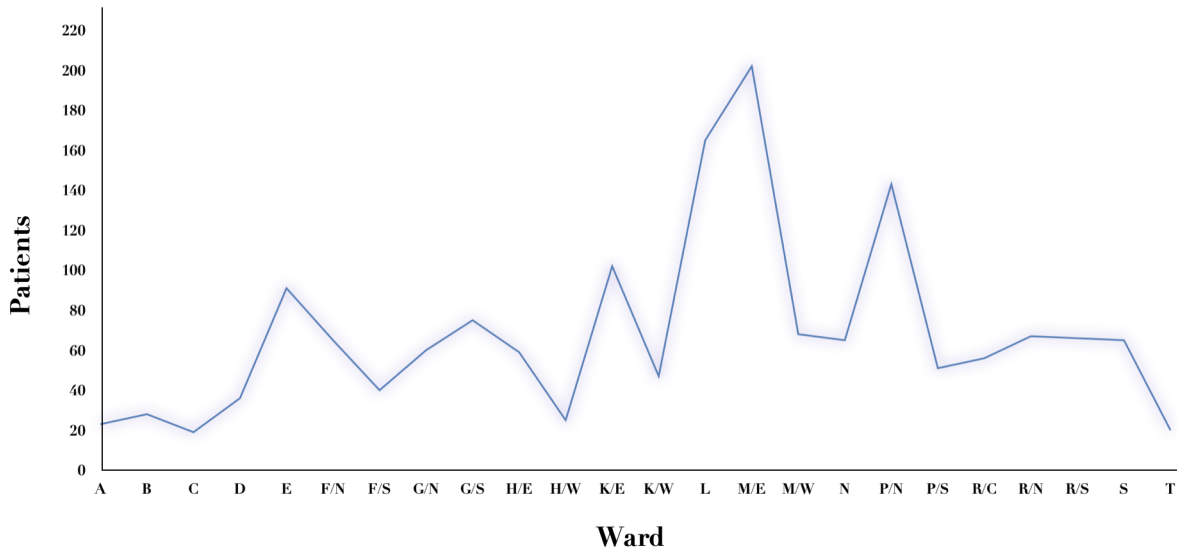


Figure 2.5: Ward Wise Statistics

The line graph presented here illustrates the ward-wise statistics derived from the BMC dataset, offering a visual representation of patient distribution across the various wards in Mumbai. This graph effectively highlights the concentration of patients in different wards, providing a clear picture of how the infection is spread throughout the city. By examining the data displayed, one can quickly grasp the relative number of patients in each ward, enabling a detailed understanding of the areas most affected by the infection.

The skewness observed in the graph is quite revealing, indicating a higher concentration of infection in certain regions. The wards listed along the X-axis are arranged in a manner that reflects their actual consecutive locations within Mumbai. This arrangement allows for an intuitive interpretation of the infection spread pattern across adjacent wards. By simply observing the graph, one can infer how the infection is propagating through neighboring areas, providing valuable insights into the dynamics of the infection's spread across the city.

A notable feature of the graph is the formation of a "bell-shaped" curve, which indicates a higher concentration of infections around Wards L and M/E. This bell-shaped distribution suggests that these wards are the epicenters of the infection, with the intensity of the infection diminishing gradually as one moves away from these central wards. The curve effectively visualizes how the infection's impact decreases across the wards on either side, illustrating a gradient of infection intensity that fades outwards from the central points of concentration.

Hospital	Patients
Kasturba Hospital	190
J J Hospital	26
Jijamata Nagar Dispensary	19
Jijamata Health Post	15
Rajawadi Hospital	14
Cooper Hospital	12
Dr. B.R. Ambedkar Hospital	12
Dr. Prashant Gandhi Hospital	12
Dr. Rupesh Mandodkar Hospital	12
Marol Health Post	12
Malvani Health Post	11
D.R. Shinde's Children Hospital	10
Gov General Hospital	10
M.W. Desai Hospital	10
Mankhurd Health Post	10
Maruti Lane Dispensary	10
Subhash Nagar Health Post	10
Others	892

Figure 2.6: Hosptial Wise Statistics

The table presented here displays hospital-wise statistics inferred from the BMC dataset. This table lists the number of individuals reported to be infected by each hospital, focusing only on those hospitals that had at least ten reported cases. Hospitals with fewer than ten reported cases have been grouped into a category labeled "Others." This approach helps in highlighting the hospitals that are significant contributors to the reported infection cases, thereby providing a clearer picture of the distribution of infected individuals across various healthcare facilities in Mumbai.

The graphs reviewed thus far have illustrated some of the most critical aspects that can be inferred from the dataset. However, there are additional modalities that warrant further exploration. These include statistics segmented by pin code, which can reveal localized hotspots of infection, and data on travel frequency via public transportation modes such as railways and the BEST buses. Since the dataset indicates a significant number of infections among infants and school-going children, mapping school locations could provide deeper insights into the primary hubs of infection spread. This multifaceted analysis will allow for a more comprehensive understanding of the factors driving the spread of Measles within the city.

Exploring these additional dimensions is crucial for gaining a holistic view of the infection dynamics. Pin code-wise statistics can help identify specific areas that require targeted intervention. Analyzing travel patterns via public transportation can shed light on how the infection spreads across different parts of the city. Furthermore, tracking school locations in relation to infection cases can reveal potential clusters of infection among young children, guiding public health officials in implementing more effective prevention and control measures. By delving into these various aspects, the study aims to uncover a detailed and nuanced understanding of Measles transmission in Mumbai.

Snapshot of the Dataset

To provide a concrete illustration of the dataset's structure, we have curated a comma-separated values (CSV) file. As depicted in the following image, this file exemplifies the data format used in our study. Each row in the CSV file represents a patient record, with columns designated for various attributes such as Patient Name, Address/Village, Hospital Registered With, Ward, and Measles Y/N. This layout ensures that each patient's information is systematically organized, allowing for efficient data processing and analysis. The illustrative snapshot of the CSV file offers a valuable window into the organization and content of the larger dataset. By showcasing this structure, we aim to facilitate a clearer understanding of the data utilized in our analysis, highlighting how each piece of information contributes to the overall investigation of Measles spread within Mumbai.



Name of Patient	Address/Village	Hospital Registered With	Ward	Measles Y/N
Person 1	Colaba	JJ Hospital	A	Y
Person 2	Antop Hill	Cooper Hospital	B	N
Person 3	Govandi	Kasturba Hospital	M	Y
Person 4	Andheri	Lilavati Hospital	L	Y

Figure 2.7: Illustration of the CSV file

Feature Vector Per Entity i.e. Hospital/Village

The study analyzed the distribution of registered cases across hospitals and villages, as illustrated in Figure 2.8. Three dummy hospitals (H1, H2, H3) and three villages (V1, V2, V3) were included in the analysis, all located at coordinates 19°E, 24°N. Hospital H1 recorded the highest number of cases at 123, followed by H2 with 45 cases, and H3 with 13 cases. Among the villages, V2 reported the most cases at 125, significantly higher than V3 with 25 cases and V1 with 20 cases. Interestingly, while hospitals collectively registered more cases (181) than villages (170), the village with the highest number of cases (V2) surpassed all individual hospitals. This data provides insights into the geographic distribution of cases and highlights potential hotspots for further investigation or resource allocation.

Hospital	# Cases	Lat/Long	Village	# Cases	Lat/Long
H1	123	19'E, 24N	V1	20	19'E, 24N
H2	45	19'E, 24N	V2	125	19'E, 24N
H3	13	19'E, 24N	V3	25	19'E, 24N

Figure 2.8: Number of Cases registered per Hospital/Village

Illustrative Graphical Representation of the Dataset

Now we show the GSP based interpretation of the Dataset thus curated. Figure 2.9 presents a GSP-based interpretation of the curated dataset, illustrating the complex relationships between hospitals and villages in the context of measles case registration. The graph G depicts six nodes: three hospitals (H1, H2, H3) in red and three villages (V1, V2, V3) in blue, interconnected to form a network. Each node is associated with a scalar value representing the number of registered measles cases, ranging from 10 to 123. The edges between nodes represent geographical relationships, with the edge weight E_{v_2,v_3} calculated using an exponential decay function based on the geographic distance between nodes. This representation allows for the analysis of spatial correlations and potential infection spread patterns. The graph structure provides a foundation for applying advanced GSP techniques to uncover hidden patterns and relationships within the measles outbreak data, potentially leading to more effective prevention and control strategies.

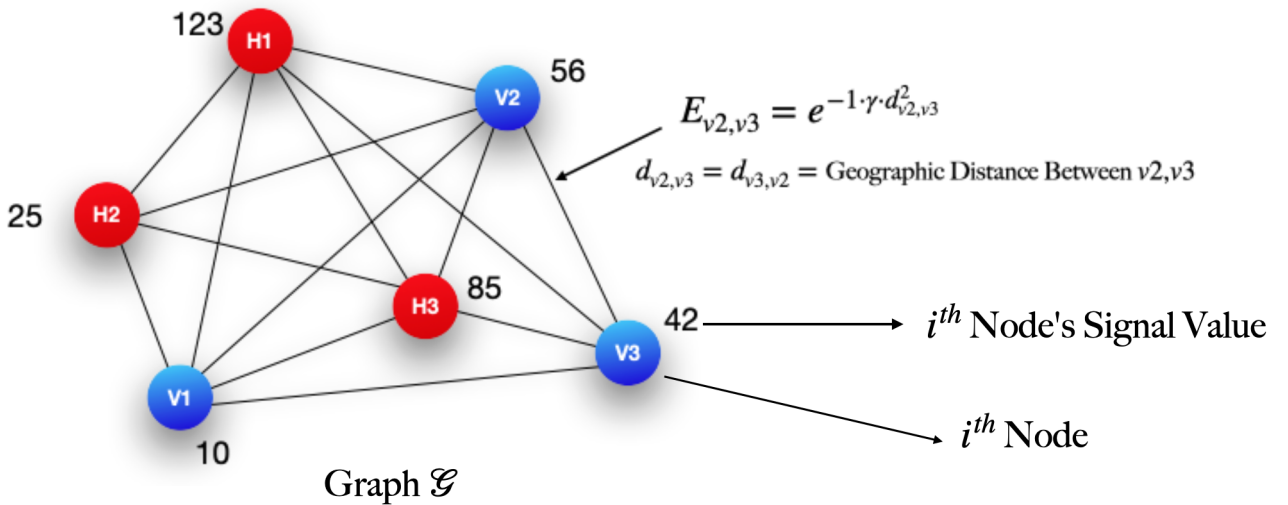


Figure 2.9: GSP-Based Interpretation of the Curated Dataset

2.3 Proposed Graph-Based Compressed Sensing Model

We address the challenge of estimating the total number of registered cases across all entities (villages and hospitals) given limited available information. This scenario aligns well with the sparse signal reconstruction framework, specifically the Graph-Based Compressed Sensing Model developed earlier in this thesis.

Let Y represent the known case counts for a subset of locations, and C denote the sensing matrix that indicates which locations have available data. Our objective is to reconstruct the complete vector of case counts across all regions using this partial information.

We approach this problem in two primary steps:

2.3.1 Sparse Representation in the Fourier Domain

First, we obtain a sparse representation of the entire case count vector in the graph Fourier space. This is formulated as an optimization problem:

$$\min_s \|Y - C \cdot \mathcal{V} \cdot s\|_2^2 + \lambda \cdot \|s\|_1$$

where $\mathcal{V}$ is the matrix of graph Fourier basis vectors, s is the sparse coefficient vector we aim to estimate, and λ is a regularization parameter that controls the sparsity of the solution. The ℓ_1 -norm penalty encourages sparsity in the Fourier domain representation.

The sensing matrix C is carefully designed to reflect the subset of locations with known data. The optimization problem is then solved to obtain the optimal sparse representation s^* . The hyperparameter λ is determined through cross-validation to ensure optimal performance.

2.3.2 Signal Reconstruction

Once we have obtained the sparse Fourier domain representation s^* , we can reconstruct the complete case count vector across all regions:

$$X^* = \mathcal{V} \cdot s^*$$

This reconstruction provides estimates for case counts in all locations, leveraging the graph structure and sparsity assumptions to infer values for regions with missing data.

This Graph-Based Compressed Sensing approach allows us to effectively estimate the complete distribution of measles cases across the entire network of hospitals and villages, even with limited initial data. The method exploits the underlying graph structure of the geographic relationships between entities, potentially revealing important spatial patterns in disease spread. Reconstruction obtained.

2.3.3 Overview and Remarks on the Proposed Model

1. This Graph-Based Compressed Sensing model offers a powerful approach for reconstructing complete epidemic data from limited observations.
2. By leveraging the underlying graph structure of geographic relationships between entities, it can infer missing case counts and potentially reveal spatial patterns in disease spread.
3. The model's ability to work with sparse data makes it particularly valuable in scenarios where comprehensive surveillance is challenging, potentially improving public health response strategies in resource-limited settings.

2.4 Choice of Sensing Matrix: Subset Selection Strategies

The performance of our model hinges significantly on the selection of subsets, making the subset selection matrix a critical component deserving special attention. For the specific tasks at hand, employing one-hot encoded matrices has demonstrated efficacy. Each row in these matrices represents a distinct subset selection method, illustrating how subsets are chosen for analysis and validation within our framework. This approach ensures that our model evaluation is robust and reflective of various subset configurations, thereby enhancing the reliability and comprehensiveness of our findings in the context of our research objectives.

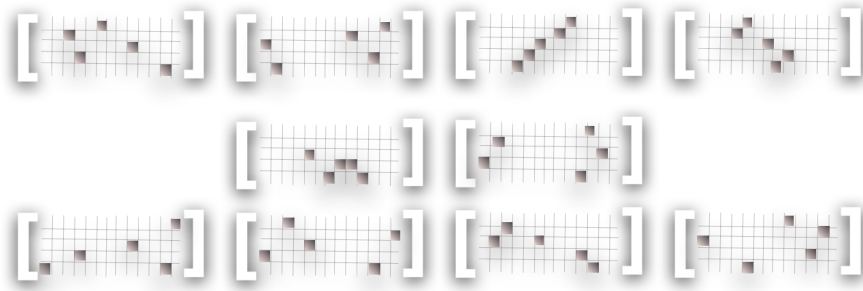


Figure 2.10: Various Subset Selector Matrices

2.4.1 Node Degree Based Subset Selection

The algorithm presented here implements a Node Degree Based Subset Selection strategy for selecting a representative subset of nodes from a graph. This approach prioritizes nodes with high degrees, meaning those connected to a larger number of other nodes in the network. The rationale behind this strategy is that high-degree nodes often serve as central or influential points within the graph structure. By selecting these nodes, the subset is likely to capture a more comprehensive representation of the overall network compared to a random selection.

The algorithm seems to function as follows:

Initialization: It begins by initializing a node degrees vector with zeros, where the length of the vector corresponds to the total number of nodes in the graph (N).

Node Degree Calculation: Then, for each node (i from 1 to N), its degree is calculated and stored in the corresponding position of the node degrees vector. **Sorting:** The node degrees vector is subsequently sorted in descending order. This ensures that nodes with the highest degrees occupy the top positions in the sorted vector.

Subset Selection: A desired subset size (m) is predetermined. The algorithm selects the first m elements from the sorted node degrees vector to create an index list containing the indices of the m highest-degree nodes.

Subset Creation: Finally, based on the index list, the algorithm creates the actual subset by selecting the corresponding nodes from the original graph

This Node Degree Based Subset Selection strategy offers a computationally efficient approach for selecting influential nodes within a network. By prioritizing highly connected nodes, it aims to capture a significant portion of the structural information encoded in the graph.

Algorithm 1 Node Degree based Subset Selection Algorithm

Initialize *node_degrees* as a vector of zeros with length N .

for $i = 1$ to N **do**

$node_degrees(i) \leftarrow D(i, i)$

end for

Sort the *node_degrees* vector in descending order, preserving the index order.

Let m be the desired subset size, e.g., $m = 500$.

Create an array *idx_mix* containing the first m elements from the sorted index vector.

Initialize a vector y with m elements, and a matrix C with m rows and N columns, all initialized to zeros.

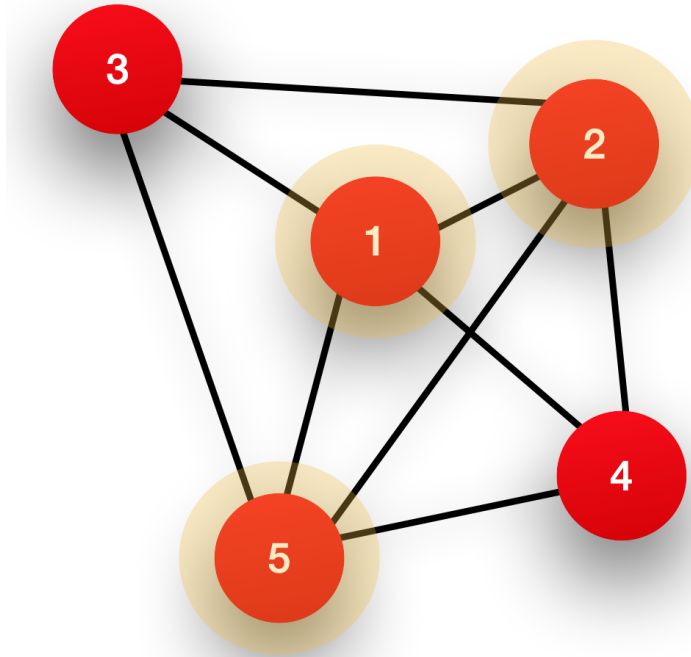
for $i = 1$ to m **do**

$y(i) \leftarrow X(idx_mix(i))$

$C(i, idx_mix(i)) \leftarrow 1$

end for

Figure 2.11: Node Degree



Node $\in \{1, 2, 5\}$ Selected
for Call of Top-3 Nodes

Figure 2.12: Illustrative Example

2.4.2 Kemeny Indicator Value Based Subset Selection

The Kemeny-based subset selection strategy integrates concepts from social choice theory into graph analysis, aiming to optimize the aggregation of individual preferences or rankings of subsets of nodes or edges within a graph. Originating from the Kemeny-Young method in voting theory, this approach entails evaluating subsets based on graph-theoretic metrics such as centrality, connectivity, and influence propagation. Each subset is assigned a preference score reflecting its contribution to these metrics, and these scores are aggregated using the Kemeny rank aggregation algorithm. This algorithm minimizes the total number of pairwise disagreements in rankings to derive a consensus ranking that prioritizes subsets according to their collective importance in achieving specific network properties or computational objectives. The theoretical foundation of the Kemeny-based strategy ensures robust and systematic subset selection, enhancing the reliability and interpretability of results in diverse graph-based applications across domains such as social networks, biological systems, and computational sciences.

$$K = \sum_{j=1}^n m_{ij} \pi_j \quad \begin{array}{l} \text{Idea Similar to Page Rank} \implies \pi = P \cdot \pi \\ m_{i,j} = \mathbb{E}[\# \text{Steps } S_i \rightarrow S_j] \\ \pi_j = \text{Stationary Distribution Probability of being in } S_j \end{array}$$

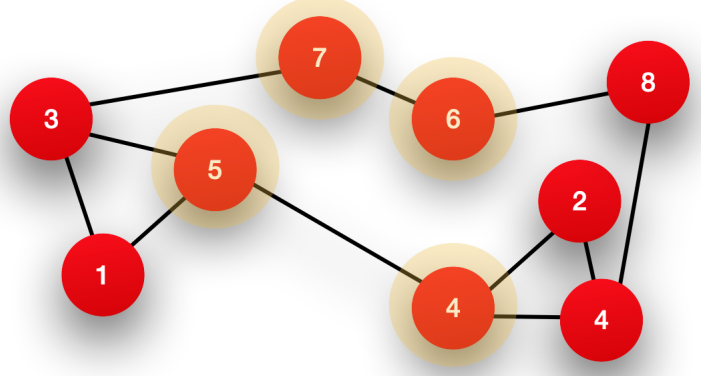
Figure 2.13: Kemeny Based Technique Connection to Page Rank Based Algorithm

Algorithm 1 Kemeny-Based Technique

- 1: **Input:** Adjacency Matrix W of size $N \times N$, K vector of zeros of length N
 - 2: **for** $i = 1$ to N **do**
 - 3: Step 1: Set Skip Node i
 - 4: Initialize A_{skip} as an $(N - 1) \times (N - 1)$ matrix from W_{-i}
 - 5: Step 2: Normalize Rows of A_{skip} to Probabilities
 - 6: Step 3: Compute Eigenvalues of A_{skip}
 - 7: Step 4: Compute Kemeny Indicator for Skip Node i
 - 8: $K(i) = 0.0$
 - 9: **for** $j = 1$ to $N - 1$ **do**
 - 10: $K(i) = K(i) + \frac{1}{1 - \lambda_j}$
 - 11: **end for**
 - 12: **end for**
 - 13: Sort the K vector in descending order, preserving the index order.
 - 14: Let m be the desired subset size, e.g., $m = 500$.
 - 15: Create an array idx_mix containing the first m elements from the sorted index vector.
 - 16: Initialize a vector y with m elements, and a matrix C with m rows and N columns, all initialized to zeros.
 - 17: **for** $i = 1$ to m **do**
 - 18: $y(i) \leftarrow X(idx_mix(i))$
 - 19: $C(i, idx_mix(i)) \leftarrow 1$
 - 20: **end for**
-

Figure 2.14: Kemeny Based Technique

The rationale of this choice is that **nodes** connecting **two communities** are associated with **very high Kemeny indicators**, as **walking** times become much **larger** if such **nodes** are removed.



Node $\in \{4,5,6,7\}$ Selected for Call of Top-4 Nodes, as these have Largest K-Value

Figure 2.15: Illustrative Example

2.4.3 Mutual Coherence Based Subset Selection

Theoretical Framework

Let $G = (V, E)$ represent the underlying graph network, where V is the set of nodes and E is the set of edges. Additionally, let F be a feature space of dimensionality N , where each node $v \in V$ is associated with a feature vector $f_v \in F$. The coherence matrix $C \in \mathbb{R}^{M \times N}$ captures the pairwise coherence between nodes, where M is the total number of nodes. Element c_{ij} of C represents the coherence between node i and node j based on their feature vectors. Here, a higher coherence value indicates a greater degree of similarity between the nodes in terms of their features.

The core objective of the algorithm is to identify a subset of m nodes ($m \ll M$) that minimizes the overall coherence within the selected set. In simpler terms, we aim to find a diverse collection of nodes that are not highly similar to each other based on the features in F . This promotes the selection of representative nodes that capture a broader spectrum of information within the network.

Technical Implementation

The algorithm iterates through each node $i \in V$. For each node, a new row vector $w_i \in \mathbb{R}^N$ is constructed as a linear combination of the rows in the feature matrix V , weighted by the corresponding elements from the i th row of the coherence matrix C . Mathematically, this can be expressed as:

$$w_i = C_i * V$$

where C_i denotes the i th row of C and $*$ represents element-wise multiplication. The maximum absolute value ($|\mu|$) of this newly created vector w_i represents the maximum coherence between

node i and the other nodes in the network based on their features.

Next, the algorithm calculates and stores the maximum coherence values ($|\mu|$) for all nodes in a vector denoted by μ . Subsequently, it sorts the μ vector in ascending order. Nodes with the lowest maximum coherence values are prioritized as they are likely less similar to other nodes in the network. The desired subset size (m) is then predefined. Finally, the algorithm extracts the top m indices from the sorted μ vector. These indices correspond to the m nodes with the lowest maximum coherence values, forming the selected subset.

Optional Extension

Some implementations might include an additional step where a separate output vector y and matrix C' are created. The vector y stores the IDs of the selected nodes, while C' stores the corresponding feature vectors for the selected nodes extracted from the original feature matrix V .

This mutual coherence based subset selection strategy offers a theoretically sound approach for selecting a diverse set of nodes from a graph network. By minimizing the overall coherence within the subset, the algorithm ensures that the selected nodes capture a wider range of information present within the network structure and feature space.

Algorithm 1 Mutual Coherence based Technique

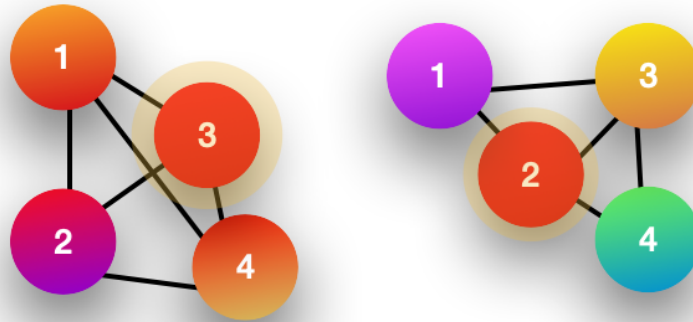
- 1: **Input:** Matrix C of size $M \times N$, Matrix V of size $N \times N$, $\vec{\mu}$ vector of zeros of length M
 - 2: **for** $i = 1$ to M **do**
 - 3: Step 1: New Row in Output Matrix := Perform Linear Combination of Rows of V weighted by weights from i^{th} row of C .
 - 4: Step 2: From this New Row, pick up the Maximum of the Absolute values and assign it to μ_i
 - 5: **end for**
 - 6: Sort the $\vec{\mu}$ vector in Ascending order, preserving the index order.
 - 7: Let m be the desired subset size, e.g., $m = 500$.
 - 8: Create an array idx_mix containing the first m elements from the sorted index vector.
 - 9: Initialize a vector y with m elements, and a matrix C' with m rows and N columns, all initialized to zeros.
 - 10: **for** $i = 1$ to m **do**
 - 11: $y(i) \leftarrow X(idx_mix(i))$
 - 12: $C'(i, idx_mix(i)) \leftarrow 1$
 - 13: **end for**
-

Figure 2.16: Mutual Coherence Based Technique

$$\mu_i = \max_j (|(C \cdot V)_{ij}|) \quad \text{for } i = 1, 2, \dots, M$$

The rationale of this choice is that nodes which have Least Largest Energy Difference with some other node, are picked up.

In Simpler terms, that Entity will be preferred to be picked up whose Neighbourhood, is most likely of similar signal values Or in some sense Less Potential Difference from Neighbourhood.



Among Node $\in \{2,3\}$, $\mu_2 \geq \mu_3$

So Node 3 will be preferred over Node 2

Figure 2.17: Illustrative Example

2.5 Hyperparameter Selection Strategy

We present a rigorous methodology aimed at accurately estimating the hyperparameter λ within the context of our study. This approach leverages a robust cross-validation framework, which systematically evaluates the performance of different λ values across multiple validation subsets derived from the dataset. By iteratively testing and validating λ through this method, we aim to identify the optimal hyperparameter that maximizes predictive accuracy or model performance, thereby enhancing the reliability and effectiveness of our computational or statistical models.

2.5.1 2-Fold Cross Validation Strategy for Hyperparameter Tuning

To determine the optimal value for the hyperparameter λ in our Graph-Based Compressed Sensing Model, we employ a 2-fold cross-validation strategy as illustrated in Figure 2.18. The process is as follows:

1. Data Preparation:

- The original data y is split into two parts: y_R (Reconstruction set) and y_V (Validation set).
- Corresponding ϕ_R and ϕ_V are constructed from the initial ϕ (representing the graph Fourier basis).

2. λ Range Definition:

- A range of λ values is defined, spanning from 10^{-9} to 10^9 .

3. Optimization Process: For each λ_i in the defined range:

- Solve the convex optimization problem using the Reconstruction set:

$$\min_{\theta} \|y_R - \phi_R \cdot \mathcal{V} \cdot \theta\|_2^2 + \lambda_i \|\theta\|_1 \quad (2.1)$$

subject to $0 \leq \mathcal{V}\theta \leq 1$

- This yields θ_{λ_i} , the optimal solution for the current λ_i .

4. Validation Error Calculation: For each λ_i :

- Compute the Validation Error:

$$VE_i = \min_{\theta} \|y_V - \phi_V \cdot \mathcal{V} \cdot \theta\|_2^2 \quad (2.2)$$

5. Optimal λ Selection:

- Choose the λ_i that produces the smallest Validation Error (VE_i) as the optimal λ .

6. Final Model:

- Use the optimal λ in the full model to obtain the final sparse representation s^* as described in the main Graph-Based Compressed Sensing Model.

This 2-fold cross-validation approach helps in selecting a λ value that balances between fitting the known data well (minimizing the reconstruction error) and maintaining sparsity in the Fourier domain (controlled by the $\lambda\|\theta\|_1$ term). By using separate reconstruction and validation sets, it ensures that the chosen λ generalizes well to unseen data, reducing the risk of overfitting.

2.5.2 Aggregate of Validation Errors across Different Measurements and Subset Selection Strategies

The presented table provides a comprehensive overview of validation errors obtained through a 2-fold cross-validation strategy aimed at tuning the hyperparameter λ . This approach seeks to identify the optimal λ value that strikes a balance between model fit on training data and its generalizability to unseen data. The table displays validation errors (VE) for various models, including Kemeny MC, Node Degree MC, and MC+Tuning, where each row likely represents a specific model configuration and each column corresponds to different λ values or indicates the minimum VE achieved.

In addition to implementing the aforementioned hyperparameter selection algorithm, we ensure diverse splitting of matrices to mitigate skewed outcomes and enhance the robustness of our results. This approach aims to minimize biases in model evaluation by varying subset selection strategies, thereby facilitating a more comprehensive assessment of model performance across different datasets and conditions.

Kemeny + MC + Tuning Lambda with Adjusted Cross Validation Subset

Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	10	6210.7335	0.3515	1	6406.0586	0.2551	195.33	0.10
300	10	6227.8715	0.3317	1	6408.6848	0.2306	180.81	0.10
350	10	6213.6477	0.3223	1	6422.9220	0.2106	209.27	0.11
370	10	6274.5631	0.3183	1	6333.8411	0.2026	59.28	0.12
400	1	6471.0417	0.1775	0.1	6603.6012	0.1745	132.56	0.00
450	1	6375.1552	0.1583	1E-09	6592.0061	0.1510	216.85	0.01
500	1E-08	6332.5379	0.1000	1E-09	6338.8620	0.0999	6.32	0.00
AVERAGE			0.2514			0.1892	142.92	0.06

Lambda Chosen which
Minimises **Cross Validation Error**

Lambda Chosen which
Minimises **RMSE**

Difference due to
CHOICE of Lambda

Node Degree + MC + Tuning Lambda with Adjusted Cross Validation Subset

Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	1E-07	7436.8451	0.2599	1	7527.4206	0.2553	90.58	0.00
300	1E-09	7394.9857	0.2348	1	7504.9050	0.2297	109.92	0.01
350	10	7557.9258	0.3227	1	7634.3165	0.2099	76.39	0.11
370	1E-09	7330.3096	0.2047	1	7408.8402	0.2033	78.53	0.00
400	1E-09	7275.3508	0.1737	0.01	7327.7095	0.1735	52.36	0.00
450	1	7423.3918	0.1588	1E-09	7461.1783	0.1521	37.79	0.01
500	1E-09	7286.5961	0.0998	1E-09	7286.5961	0.0998	0.00	0.00
AVERAGE			0.2078			0.1891	63.65	0.02

Lambda Chosen which
Minimises **Cross Validation Error**

Lambda Chosen which
Minimises **RMSE**

Difference due to
CHOICE of Lambda

MC + Tuning Lambda with Adjusted Cross Validation Subset

Measurements	Lambda @ min VE	Validation Error VE	RMSE @ Lambda	Lambda @ min RMSE	Validation Error VE	RMSE	VE Difference	RMSE Difference
250	10	3007.4391	0.3494	1	3037.8568	0.2559	30.42	0.09
300	10	3022.3485	0.3316	1	3068.4093	0.2283	46.06	0.10
350	10	3041.0553	0.3219	1	3145.0702	0.2098	104.01	0.11
370	10	3113.4056	0.3184	0.1	3374.6716	0.1968	261.27	0.12
400	10	3438.7900	0.3153	0.1	3655.5488	0.1726	216.76	0.14
450	10	3436.4082	0.3104	1E-09	3632.9500	0.1510	196.54	0.16
500	10	3500.3987	0.3031	1E-09	3706.7796	0.0999	206.38	0.20
AVERAGE			0.3214			0.1877	151.63	0.13

Lambda Chosen which
Minimises **Cross Validation Error**

Lambda Chosen which
Minimises **RMSE**

Difference due to
CHOICE of Lambda

Figure 2.18: Hyperparameter Selection Strategy

2.6 Results obtained from the Proposed Model

2.6.1 BMC-Measles: Node Granularity H+V

Measurements	Random	Node Degree	Kemeny	MC	AC	Node Degree + MC	Kemeny + MC	Node Degree + AC	Kemeny + AC	True Greedy MC
250	0.9222	0.4568	0.8407	0.2559	0.9037	0.2605	0.2595	0.9048	0.9078	0.9154
300	0.3267	0.3991	0.8334	0.2283	0.9071	0.2297	0.2371	0.9068	0.9046	0.2840
350	0.8891	0.3878	0.8281	0.2137	0.9054	0.2135	0.2159	0.9050	0.9011	0.2791
370	0.2498	0.3806	0.1255	0.1975	0.3865	0.2047	0.2066	0.9034	0.9007	0.2691
400	0.8310	0.3744	0.1145	0.1728	0.3625	0.1739	0.1748	0.3625	0.3618	0.2231
450	0.2639	0.3648	0.0881	0.1513	0.3483	0.1524	0.1512	0.3482	0.3497	0.2221
500	0.1348	0.3403	0.0481	0.1000	0.3222	0.0999	0.0999	0.3222	0.3222	0.1751

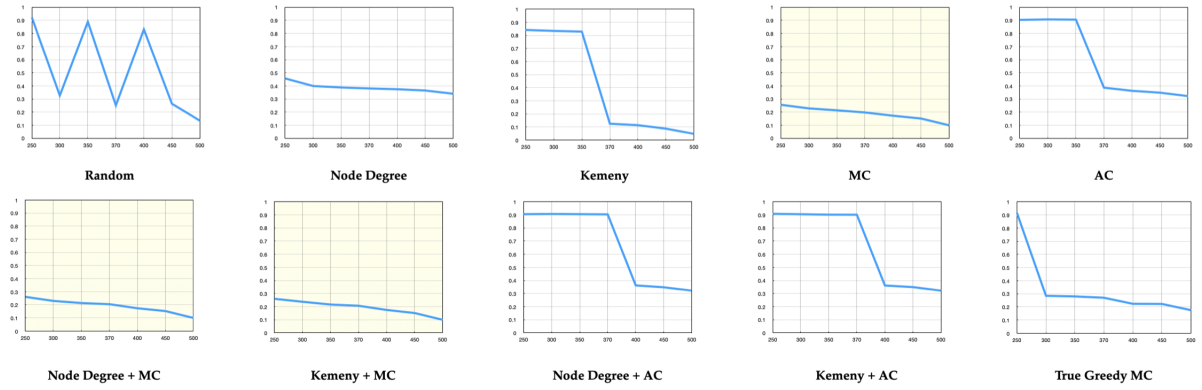


Figure 2.19: BMC-Measles: Node Granularity H+V

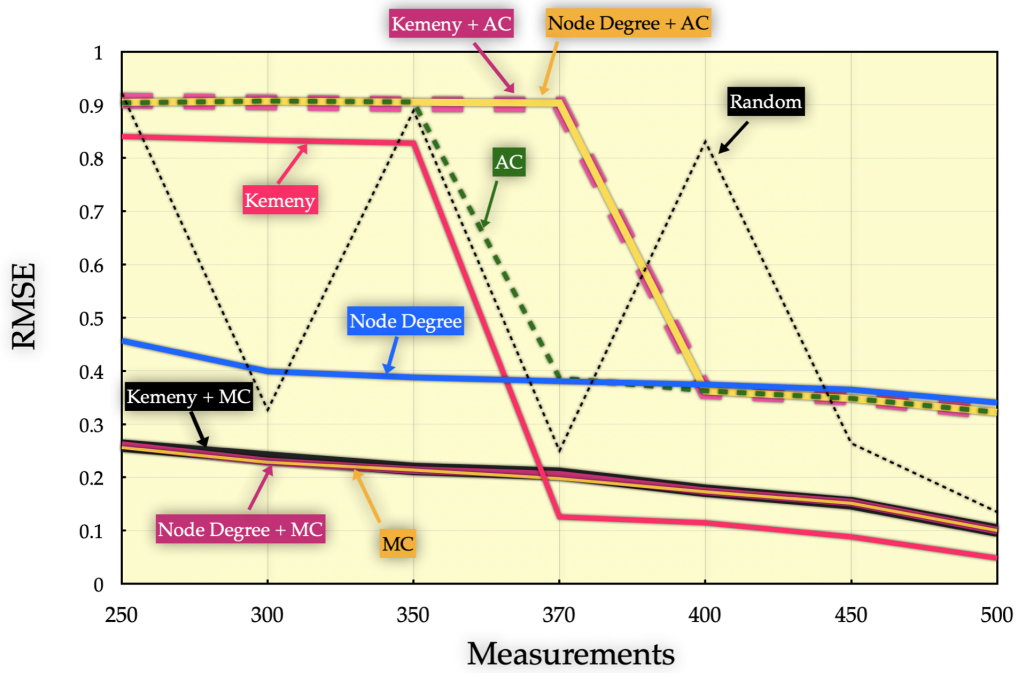


Figure 2.20: Comparison of all Techniques presented in the above table wrt RMSE

Random					MC				
Measurements	MSE	Min Difference	Max Difference	Avg. Difference	Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	56.0792	7.269E-11	154.7108	1.1086	250	4.3173	5.748E-10	16.0000	1.0851
300	7.0395	5.794E-11	38.9384	0.8353	300	3.4380	4.160E-10	16.0000	0.8466
350	52.1276	2.354E-11	154.2275	0.8365	350	3.0123	1.088E-10	15.9890	0.5530
370	4.1162	3.357E-09	18.5366	0.8109	370	2.5733	4.470E-11	15.9340	0.4778
400	45.5331	5.119E-11	154.4227	0.5562	400	1.9685	1.958E-11	15.8753	0.3929
450	4.5915	1.150E-10	38.9836	0.3266	450	1.5092	7.733E-09	16.0000	0.2423
500	1.1983	1.481E-10	20.7327	0.1271	500	0.6598	8.802E-09	13.0000	0.0993

Node Degree					Node Degree + MC				
Measurements	MSE	Min Difference	Max Difference	Avg. Difference	Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	13.7607	1.536E-10	43.9872	1.4186	250	4.4734	7.420E-11	15.9927	0.9654
300	10.5033	2.725E-10	43.9805	0.9841	300	3.4785	6.051E-10	16.0000	0.8646
350	9.9197	2.400E-09	44.0000	0.8449	350	3.0063	1.314E-10	15.9127	0.5614
370	9.5501	8.988E-09	44.0000	0.7339	370	2.7629	4.019E-11	15.9416	0.5043
400	9.2461	3.810E-11	43.9908	0.6113	400	1.9935	2.570E-09	15.9958	0.3979
450	8.7739	1.561E-09	44.0000	0.7575	450	1.5325	1.582E-08	16.0000	0.2537
500	7.6384	2.460E-12	43.4076	0.3191	500	0.6575	1.176E-09	13.0000	0.0969

Kemeny					Kemeny + MC				
Measurements	MSE	Min Difference	Max Difference	Avg. Difference	Measurements	MSE	Min Difference	Max Difference	Avg. Difference
250	46.6021	6.780E-11	154.2739	0.9199	250	4.4407	7.121E-11	15.8394	0.9356
300	45.7964	4.694E-12	154.0280	0.7425	300	3.7075	2.030E-12	15.9926	0.6970
350	45.2192	5.520E-11	154.1698	0.6026	350	3.0742	9.500E-12	15.8912	0.5666
370	1.0389	2.961E-09	8.0611	0.3615	370	2.8160	2.410E-08	16.0000	0.5077
400	0.8639	1.292E-10	8.0480	0.2873	400	2.0145	1.185E-10	15.8651	0.4095
450	0.5117	2.170E-08	5.4665	0.1853	450	1.5081	1.343E-08	16.0000	0.2422
500	0.1529	7.730E-12	5.4924	0.0605	500	0.6581	1.404E-10	12.9934	0.0993

Figure 2.21: Comparison of all Techniques presented in the above table wrt MSE

2.6.2 BMC-Measles: Node Granularity H

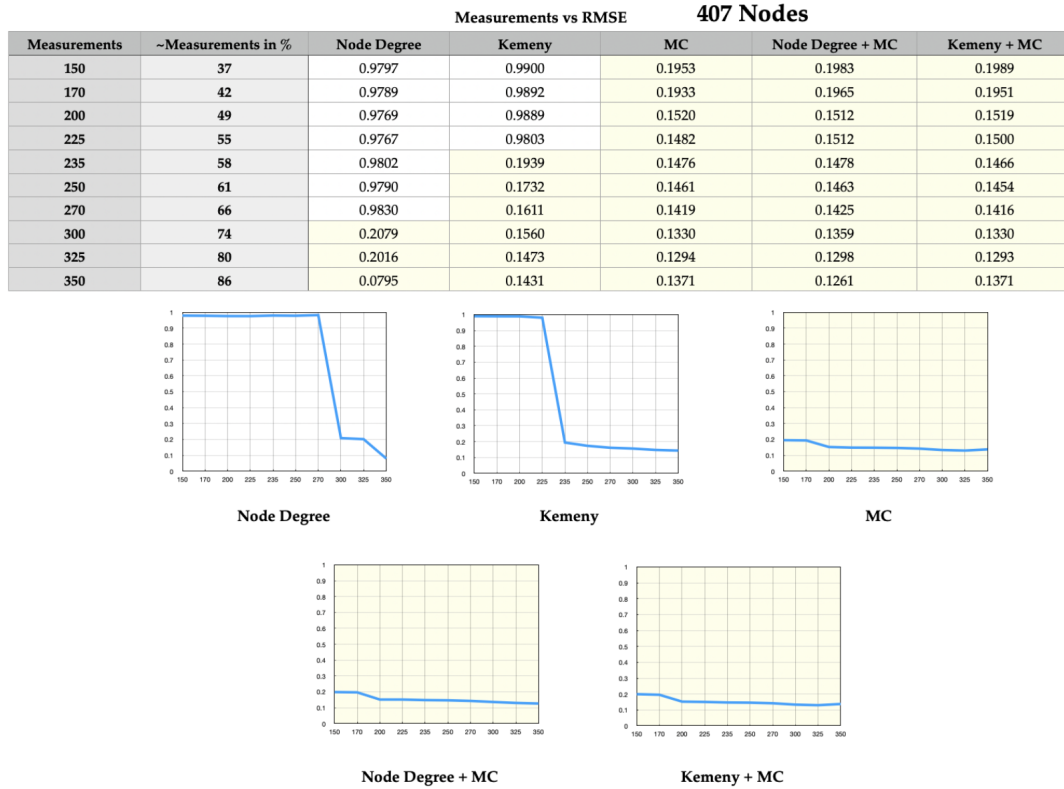


Figure 2.22: BMC-Measles: Node Granularity H

2.6.3 BMC-Measles: Node Granularity V

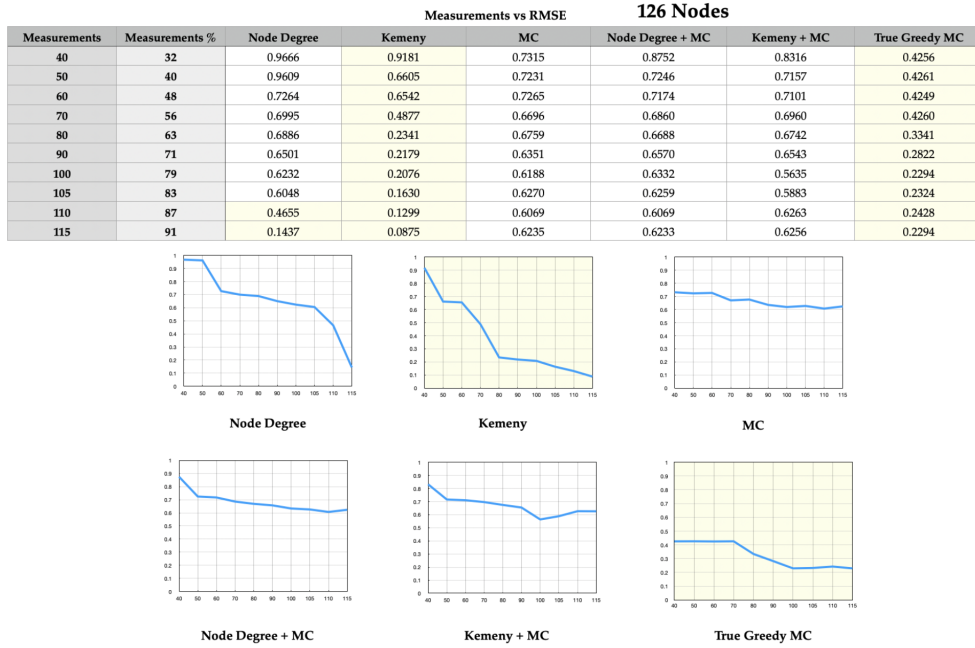


Figure 2.23: BMC-Measles: Node Granularity V

2.7 Case Study-1: Analysis of the Measles Outbreak Over the Data Procured from WHO

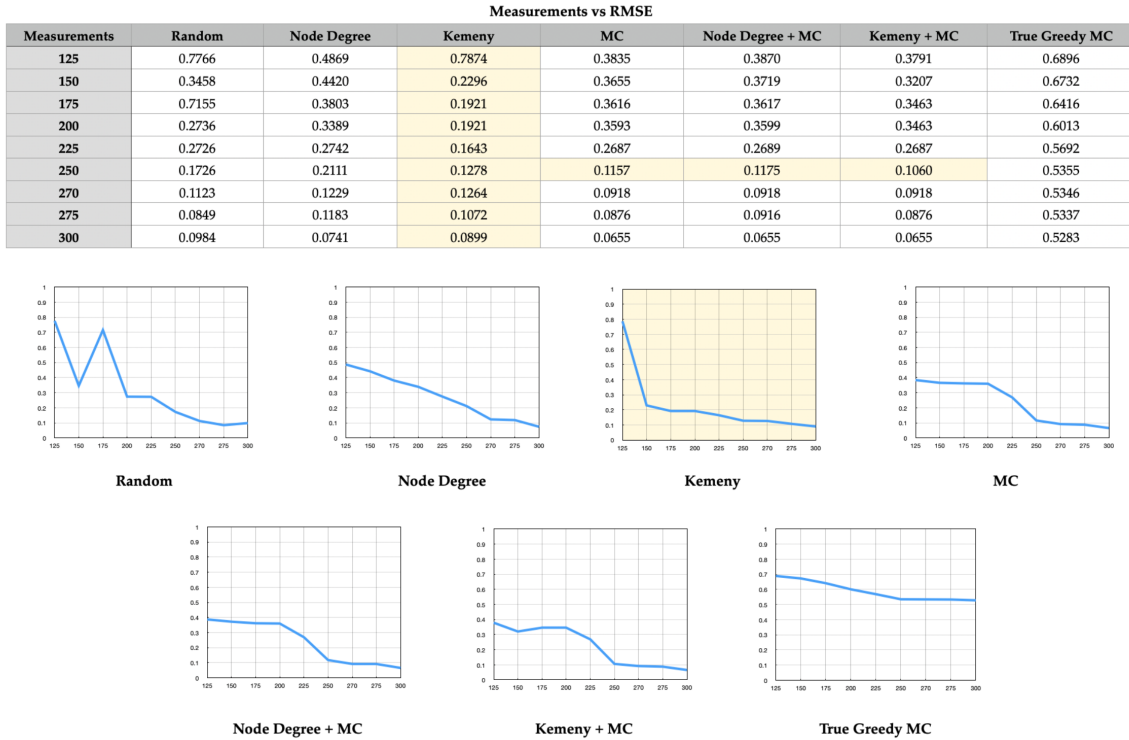


Figure 2.24: WHO-Measles: Node Granularity H+V

Measurements vs MSE							
Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
125	3330.92	1309.60	3424.24	812.14	827.18	793.61	2626.57
150	660.26	1078.76	2994.78	737.87	763.99	567.97	2503.15
175	2827.24	798.96	291.04	722.03	722.38	662.24	2273.32
200	413.51	634.15	203.89	713.09	715.24	662.28	1996.65
225	410.38	415.36	149.08	398.79	399.34	398.81	1789.32
250	164.49	246.23	90.23	73.93	76.26	62.08	1583.54
270	69.70	83.43	88.20	46.52	46.53	46.52	1578.44
275	39.76	77.28	63.51	42.40	46.38	42.40	1573.34
300	53.52	30.33	44.61	23.67	23.67	23.67	1541.75

Figure 2.25: Comparison of all the Techniques against MSE

Measurements vs RMSE							
Measurements	Random	Node Degree	Kemeny	MC	Node Degree + MC	Kemeny + MC	True Greedy MC
12	0.9305	0.9214	0.4023	0.2233	0.3135	0.2233	0.9453
14	0.3532	0.8771	0.3284	0.2103	0.2296	0.2103	0.9415
16	0.9436	0.3830	0.2862	0.1989	0.2080	0.1989	0.9626
18	0.2557	0.2389	0.1876	0.1054	0.2027	0.1054	0.1030
20	0.1897	0.1664	0.1536	0.1084	0.0596	0.1084	0.1321

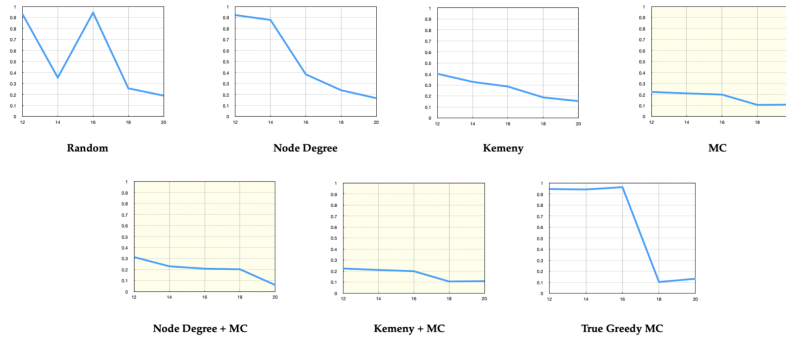


Figure 2.26: WHO-Measles: Ward Granularity

2.8 Case Study-2: Graph Frequency Analysis of Covid-19 Incidence to identify County-Level Contagion Patterns in the US

2.8.1 Dataset Details

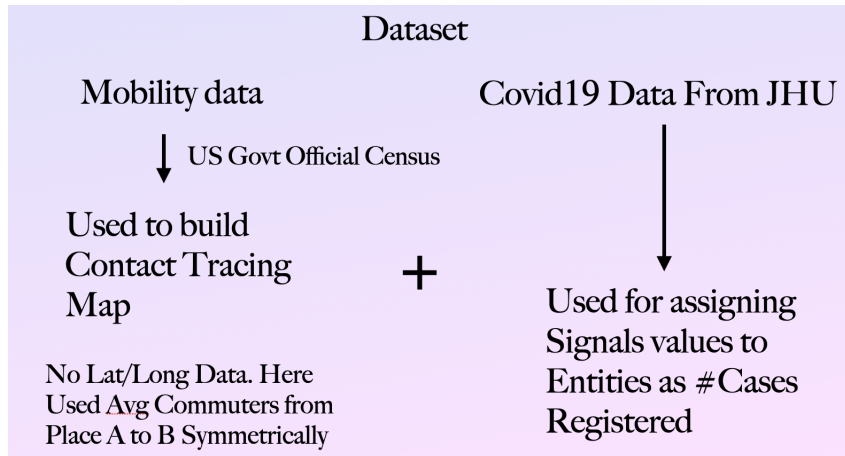


Figure 2.27: Model Setup and Dataset Details

2.8.2 US County: Province Granularity

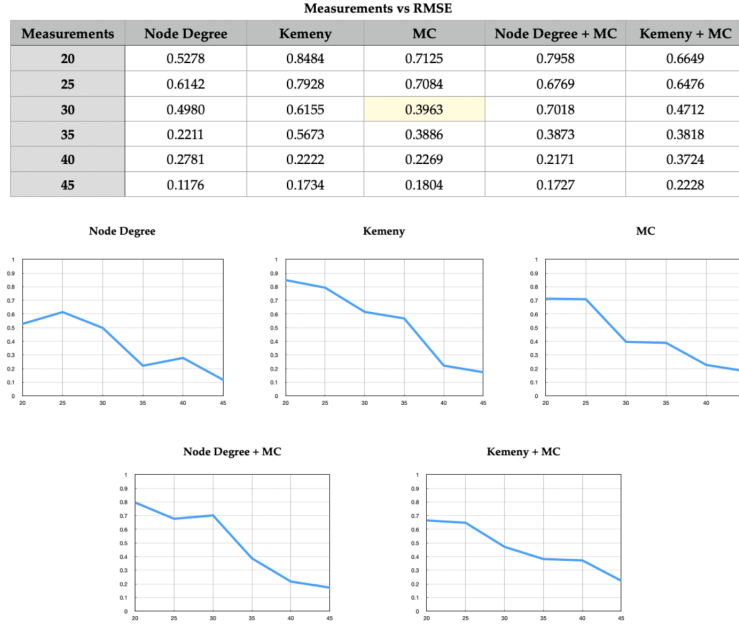


Figure 2.28: US-County Province Granularity

2.8.3 US County: Province Granularity with Log Scale Mobility Data

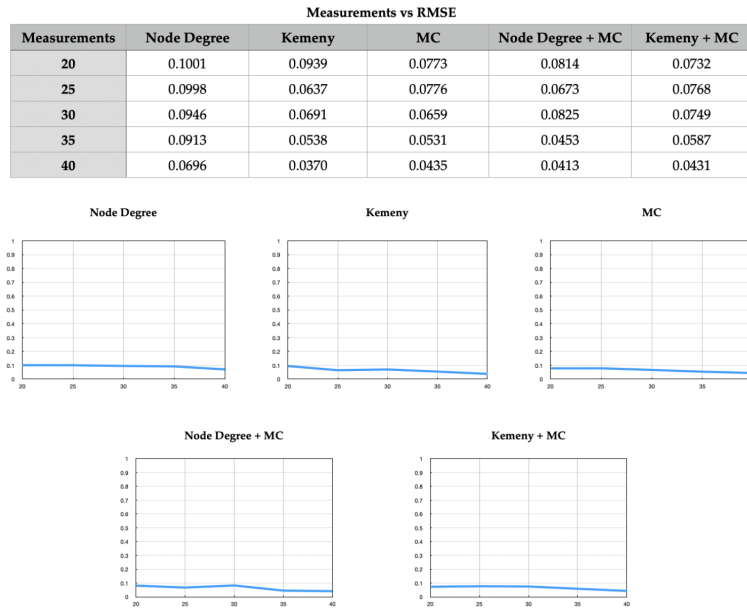


Figure 2.29: Province Granularity (Log Scale Mobility Data)

Chapter 3

Graph Learning

3.1 Problems in the Proposed Graph-Based Compressed Sensing Model

It is succinct that the success of the Graph-Based Compressed Sensing model in achieving optimal reconstruction of properties within a population is intricately tied to the quality of the underlying graph structure employed. The efficacy of this model hinges upon how accurately the graph captures the relationships and dependencies between entities or variables under study.

A high-quality graph, characterized by its ability to faithfully represent the true underlying interactions or similarities, facilitates more accurate inference and reconstruction of the latent properties of the population. Conversely, a poorly constructed graph may introduce biases or inaccuracies, limiting the model's ability to effectively utilize the available data for reconstruction purposes. Therefore, the careful construction and selection of the graph play a pivotal role in determining the overall performance and reliability of the Graph-Based Compressed Sensing approach in practical applications.

When the edge weights are not naturally defined by an application, one common way to define the weight of an edge connecting vertices i and j , $W_{i,j}$, is via a thresholded Gaussian kernel weighting function.

$$W_{i,j} = \begin{cases} \exp\left(-\frac{[\text{dist}(i, j)]^2}{2\theta^2}\right) & \text{if } \text{dist}(i, j) \leq \kappa \\ 0 & \text{otherwise,} \end{cases}$$

Figure 3.1: Gaussian RBF function being chosen as Common Edge-weighting heuristic

For some parameters θ and κ , $\text{dist}(i, j)$ may represent a physical distance between vertices i and j , or the Euclidean distance between two feature vectors describing i and j . This is a heuristic that is commonly applied to represent the edge weights, thus we wished to infer directly from the data.

When a natural choice of the graph is not readily available from the data sets, it is thus desirable to infer or learn a graph topology from the data.

3.2 Graph Learning: History, Development, Impact, and Outlook

The graph-learning problem is of significant importance due to several key reasons:

1. **Capturing Data Geometry:** Graphs can effectively capture the intrinsic geometry or structure of complex data. This capability is crucial for efficient data processing, analysis, and visualization tasks, enabling insights into the underlying relationships and patterns that may not be apparent in raw data alone.
2. **Learning Entity Relationships:** Learning the relationships between entities in data is beneficial across various domains. For instance, in neuroscience, understanding the functional connectivity between different brain regions aids in studying brain networks and cognitive processes. Similarly, in social network analysis, identifying behavioral influences between individuals or groups can provide insights into social dynamics and influence propagation.
3. **Predictive Power:** The inferred graph structure can be leveraged to predict future data evolution. By modeling dependencies and interactions between entities, predictive models can anticipate how the system or network of interest will evolve over time. This predictive capability is particularly valuable in forecasting trends, optimizing resource allocation, and making informed decisions based on anticipated changes.

3.2.1 What is Graph Learning?

Graph learning is a fundamental problem in data science and machine learning that focuses on inferring the structure of a graph from data. This problem is essential in various fields, such as neuroscience, social network analysis, and biology, where understanding the relationships among entities is crucial for insights and decision-making.

Problem Formulation: Given T observations for N entities, we aim to infer the underlying relationships that exist among these entities. These observations are typically organized in a matrix where each column represents a graph signal defined over all entities. For example, in a social network, the entities could be individuals, and the observations could be different attributes or behaviors measured over time.

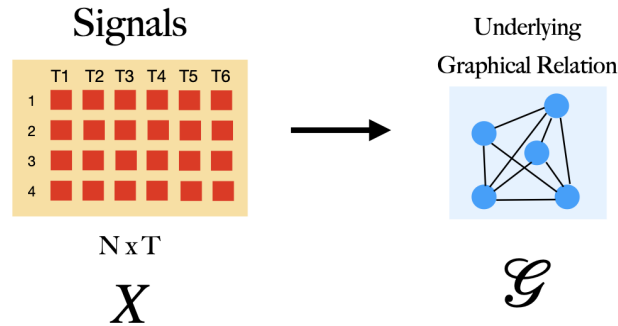


Figure 3.2: Graph Learning

The process involves analyzing these observations to construct a graph that accurately reflects the relationships and interactions between the entities. In this graph, the nodes correspond to

the entities, and the edges capture the strength or nature of the inferred relationships. The quality of the inferred graph is crucial, as it impacts the effectiveness of subsequent analyses and applications, such as clustering, classification, and anomaly detection.

Ultimate Objective:

The ultimate objective of graph learning is to construct a graph that accurately represents the underlying relationships and interactions among entities based on observed data. This graph can then be used to gain deeper insights into the structure and dynamics of the system, enabling better understanding, prediction, and decision-making.

3.2.2 Motivating Example

In brain signal analysis, graph learning techniques are pivotal. Consider blood-oxygen-level-dependent (BOLD) signals, derived from functional magnetic resonance imaging (fMRI), capturing brain dynamics. Neuroscience focuses on inferring functional connectivity—statistical relationships among brain regions during tasks or rest, revealing communication insights.

Analyzing BOLD signals identifies synchronized activity patterns, hinting at functional connections. Graph structures visualize these connections, where nodes denote brain regions and edges quantify connectivity strength.

Inferencing the graph from a multivariate BOLD time series is critical for several reasons:

- It aids in understanding the normal functioning of the brain by revealing the network of interactions between different regions.
- It helps identify abnormal connectivity patterns associated with neurodegenerative diseases, such as Alzheimer’s or Parkinson’s disease, potentially serving as biomarkers for early diagnosis or monitoring progression.
- It provides a basis for developing targeted interventions and treatments by identifying critical regions and connections influencing disease states or cognitive functions.

Thus, inferring a graph structure from BOLD signals is not merely a technical challenge but a crucial step towards advancing our understanding of brain function and dysfunction. This example illustrates the broader implications and potential of graph learning in real-world applications, particularly in neuroscience.

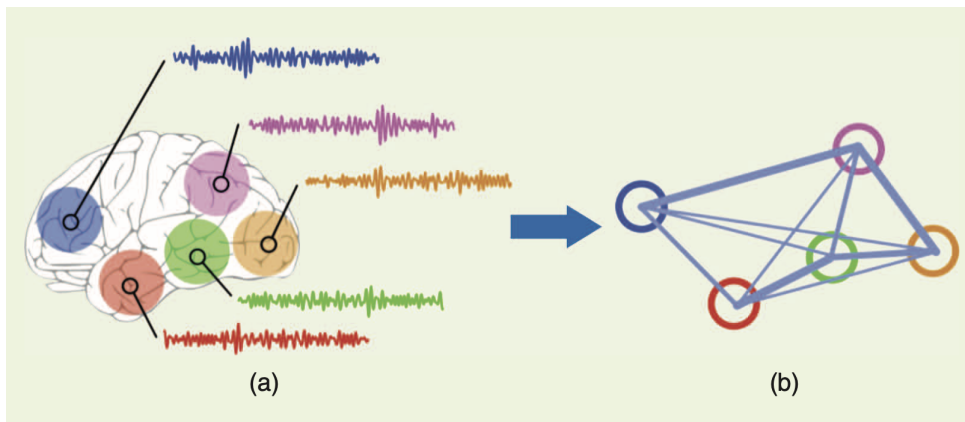


Figure 3.3: Illustrative image adapted from Dong et al.: (a) BOLD time series data from different brain regions, and (b) a functional connectivity graph with vertices representing brain regions and edges (thicker lines indicate stronger connections) representing functional connectivity.

3.2.3 Statistical Approach to Graph Learning using Probabilistic Graphical Models

The general philosophy behind the statistical view is that there exists a graph G , whose structure determines the joint probability distribution of the observations on the data entities, i.e., columns of the data matrix X .

Such models are known as probabilistic graphical models, where the edges in the graph encode conditional independence relationships among the random variables represented by the vertices.

There are two main types of graphical models:

1. **Undirected graphical models**, also known as Markov random fields (MRFs), in which local neighborhoods of the graph capture the independent structure of the variables.
2. **Directed graphical models**, also known as Bayesian networks or belief networks (BNs), which have a more complex notion of independence by considering the direction of edges.

Both MRFs and BNs have their respective advantages and disadvantages. In this section, we focus primarily on the approaches for learning MRFs, which offer a simpler representation of conditional independence and have connections to graph signal processing (GSP)-based methods.

How in Graphical Models, Edge Weights Were Inferred?

Consider the Following Toy Setting. Suppose we have 4 random variables, $X = \{X_1, X_2, X_3, X_4\}$, and below is the graphical representation of these variables, associated with edges as shown:

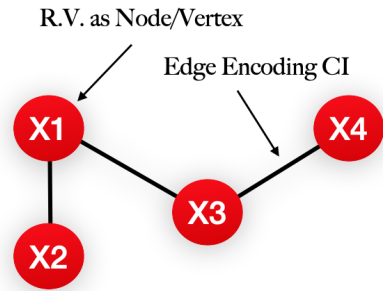


Figure 3.4: Toy Graphical Model

As per PGMs, the joint probability distribution over a set of random variables depicted using a graphical model would be defined as:

$$P(X) \propto \prod_{i=1}^N \phi_i \cdot \prod_{i,j=1, i \neq j}^N \phi_{i,j}$$

For an exponential family of distributions:

$$\phi_i = e^{\theta_{i,i} \cdot x_{i,i}^2}, \quad \phi_{i,j} = e^{\theta_{i,j} \cdot x_i \cdot x_j}$$

The joint probability distribution now becomes:

$$P(X) = \frac{1}{Z(\theta)} \cdot e^{\sum \theta_{i,i} \cdot x_{i,i}^2 + \sum_{i \neq j} \theta_{i,j} \cdot x_i \cdot x_j}$$

Finding the parameter matrix Θ was a way to infer the relation among nodes in graphical models:

$$P(X) = \frac{1}{Z(\theta)} \cdot e^{\sum \theta_{i,i} \cdot x_{i,i}^2 + \sum_{i \neq j} \theta_{i,j} \cdot x_i \cdot x_j}$$

But we know, the closed-form joint probability distribution of the Gaussian Graphical Model (GMRF) with $\mu = 0$:

$$p(\mathbf{x}|\mathbf{\Theta}) = \frac{|\mathbf{\Theta}|^{1/2}}{(2\pi)^{N/2}} \exp\left(-\frac{1}{2} \mathbf{x}^T \mathbf{\Theta} \mathbf{x}\right)$$

Figure 3.5: Closed Form Probability Distribution over GMRF

So the way to proceed from here on was:

$$P(x) = \frac{|\Theta|^{1/2}}{(2\pi)^{N/2}} \cdot e^{-\frac{1}{2} x' \Theta x}$$

The log-likelihood is given by:

$$\log(L(X)) = \sum_{i=1}^N \log(P(x_i))$$

We then maximize the log-likelihood with respect to Θ :

$$\max_{\Theta} \log(L(X))$$

Following is what we obtain on solving the above Log-Likelihood problem,

$$\begin{aligned} \underset{\mathbf{\Theta}, \sigma^2}{\text{maximize}} \quad & \log \det \mathbf{\Theta} - \text{tr}\left(\frac{1}{M} \mathbf{X} \mathbf{X}^T \mathbf{\Theta}\right) - \rho \|\mathbf{\Theta}\|_1, \\ \text{subject to} \quad & \mathbf{\Theta} = \mathbf{L} + \frac{1}{\sigma^2} \mathbf{I}, \mathbf{L} \in \mathcal{L}, \end{aligned}$$

Figure 3.6: Graphical LASSO

Note:

1. It was desired to learn those kinds of inverse covariance matrices, where off-diagonal elements were ≤ 0 . Ensuring this in the search space, as evident from the above, the authors obtain the Graph Laplacian Matrix, or equivalently, an adjacency matrix with non-negative weights.
2. The trace term in the above includes the so-called Laplacian quadratic form $X^T L X$, which measures the smoothness of the data on the graph and has also been used in other approaches that are not necessarily developed from the viewpoint of inverse covariance estimation.

3.2.4 Graph Learning using Graph Signal Processing: Evolution of PGMs

There is a growing interest in the signal processing community to analyze signals that are supported on the vertex set of weighted graphs, leading to the fast-growing field of GSP. GSP enables the processing and analysis of signals that lie on structured but irregular domains by generalizing classical signal processing concepts, tools, and methods, such as time-frequency analysis and filtering, on graphs. Consider a weighted graph $G = \{V, E\}$ with the vertex set V of cardinality N and edge set E . A graph signal is defined as a function $x : V \rightarrow \mathbb{R}^N$ that assigns a scalar value to each vertex. When the graph is undirected, the combinatorial or unnormalized graph Laplacian matrix L is defined as

$$L = D - W,$$

where D is the degree matrix that contains the degrees of the vertices along the diagonal, and W is the weighted adjacency matrix of G . Since L is a real and symmetric matrix, it admits a complete set of orthonormal eigenvectors with the associated eigenvalues via the eigencomposition

$$L = \chi \Delta \chi',$$

where χ is the eigenvector matrix that contains the eigenvectors as columns, and Δ is the eigenvalue matrix $\text{diag}(\lambda_0, \lambda_1, \dots, \lambda_{N-1})$ that contains the eigenvalues along the diagonal. Conventionally, the eigenvalues are sorted in increasing order, and for a connected graph: $\lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{N-1}$. The Laplacian matrix L enables a generalization of the notion of frequency and the definition of a graph Fourier transform (GFT). Alternatively, a GFT may also be defined using the adjacency matrix W , and this definition can be used in directed graphs. Furthermore, both L and W can be interpreted as a general class of shift operators on graphs. These operators are used to represent and process signals on a graph in a way similar to that in traditional signal processing.

Dong et. al. Approach

$$\begin{aligned} \min_{L \in \mathbb{R}^{n \times n}, Y \in \mathbb{R}^{n \times p}} \quad & ||X - Y||_F^2 + \alpha \text{tr}(Y^T L Y) + \beta ||L||_F^2 \\ \text{s.t.} \quad & \text{tr}(L) = n, \\ & L_{ij} = L_{ji} \leq 0, i \neq j, \\ & L \cdot \mathbf{1} = \mathbf{0}, \end{aligned}$$

Figure 3.7: Dong et al's Formulation for Graph Learning, a GSP SOTA Technique

$$\begin{aligned} X &\in \mathbb{R}^{n \times p} \\ Y &= \chi \cdot h \quad y^T \cdot \mathcal{L} \cdot y : \text{Gives Measure of Smoothness on the Graph} \\ \dim \chi & n \times n \quad \because y^T \cdot \mathcal{L} \cdot y = \frac{1}{2} \cdot \sum_{i,j} w_{i,j} \cdot (y(i) - y(j))^2 = 2 \cdot \text{tr}(Y^T \cdot L \cdot Y) \\ \dim h & n \times p \end{aligned}$$

Figure 3.8: Notations

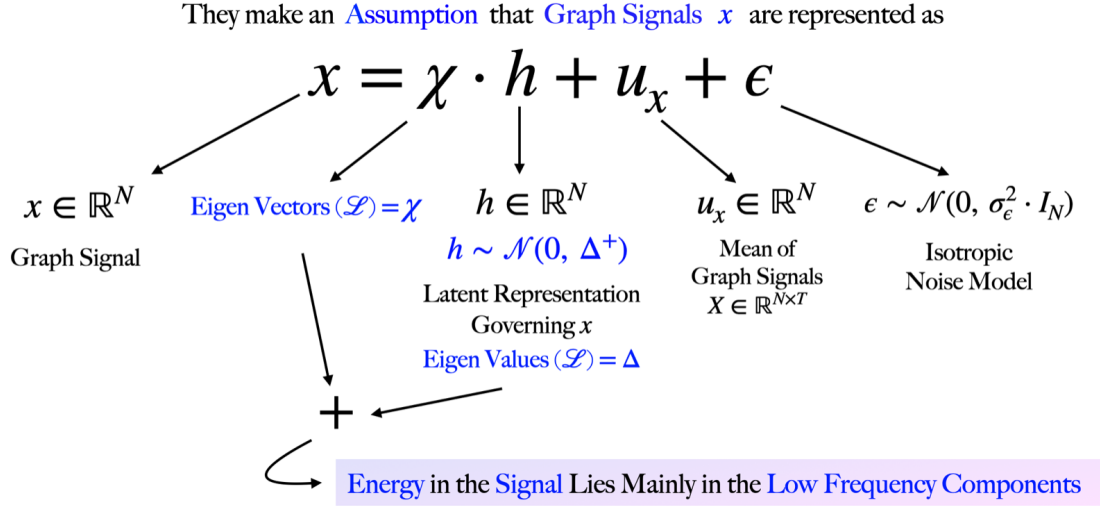


Figure 3.9: Assumption in the Graph Signal being Generated

They tackle graph Laplacian learning to infer graph topologies that best suit smooth variations in input data signals. Using a factor analysis model and Gaussian priors on latent variables, they ensure efficient representations favoring signal smoothness. Their proposed algorithm minimizes signal variations on the learned graph, enforcing this desired property effectively.

Kalofolias's Approach

For obvious reasons, a Graph Laplacian is Never an alternative to an Inverse Covariance Matrix, still its being using the Inverse Covariance Estimation problems, by imposing Complicated Constraints over the Search Space. Even in the Field of GSP, just because it is an evolved version of PGMs to tackle Graph Learning, is getting restricted to Graph Laplacians, while originally the problem's requirement is itself not a Graph Laplacian, but an Adjacency Matrix. The main idea is instead of $\mathcal{L}$, we are learning $\mathcal{W}$.

Let us define the *pairwise distances matrix* $Z \in \mathbb{R}_+^{m \times m}$ $Z_{i,j} = \|x_i - x_j\|^2$

Consider the Smoothness term: $\text{tr}(X^\top L X) = \frac{1}{2} \text{tr}(W Z) = \frac{1}{2} \|W \circ Z\|_{1,1}$

$$\underset{W \in \mathcal{W}_m}{\text{minimize}} \quad \|W \circ Z\|_{1,1} - \alpha \mathbf{1}^\top \log(W \mathbf{1}) + \beta \|W\|_F^2$$

Ensures Graph Connectedness

Imposes Graph Sparsity

Figure 3.10: Kalofolias Implementation of Graph Learning in GSP: An Adjacency Matrix POV

Humbert et. al.'s Approach

Notations

Notation	Dimension	Meaning
N	scalar	Total Number of Entities among which we wish to learn the graphical relation.
T	scalar	Total Time Steps recorded over Entities.
X	$N \times T$	Signals : T measurements taken over each of the N Entities.
V	$N \times N$	Eigen Vectors of the Graph Laplacian L .
Δ	$N \times N$	A Diagonal Matrix storing Eigen Values of the Graph Laplacian L .
α	$N \times T$	Representation of Signals in Fourier Space with Basis V also called as GFT coefficients. (Graph Fourier Transform)
W	$N \times N$	Adjacency Matrix of the underlying Graphical relation between the entities that we wish to learn.
D	$N \times N$	Diagonal Matrix storing Degrees of each Node (or Entity) in the present in the Graph. Mathematically expressed as $D_{i,i} = \sum_j W_{i,j}$.
L	$N \times N$	Combinatorial Graph Laplacian Matrix L (or Unnormalized Graph Laplacian L) corresponding to the Adjacency Matrix W . Mathematically expressed as $L = D - W$. The Eigen Decomposition of Graph Laplacian L is expressed as $L = V \cdot \Delta \cdot V^\top$
$\hat{\Theta}$	$N \times N$	Inverse Covariance Matrix Corresponding to the Signals stored in X . Mathematically expressed as $\hat{\Theta} = \left(\frac{X \cdot X^\top}{T-1}\right)^{-1}$.
σ	scalar	The Relation between the Inverse Covariance Matrix and the Combinatorial Graph Laplacian Matrix L is expressed as $\hat{\Theta} = L + \frac{1}{\sigma^2} \cdot I$, here the parameter σ , controls the amount of regularization .

$$\begin{aligned}
& \min_{\alpha, \Delta, V} \|X - V \cdot \alpha\|_F^2 + \mu_1 \cdot \|\Delta^{1/2} \cdot \alpha\|_F^2 + \mu_2 \cdot \|\alpha\|_S \\
& \text{s.t.} \quad \begin{cases} V^\top V = I_N, v_1 = \frac{1}{\sqrt{N}} \mathbf{1}_N, & \text{(a)} \\ \left(V \Delta V^\top\right)_{k,\ell} \leq 0 \quad k \neq \ell, & \text{(b)} \\ \Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \succeq 0, & \text{(c)} \\ \text{tr}(\Delta) = N \in \mathbb{R}_*^+, & \text{(d)} \end{cases} \quad (3.1)
\end{aligned}$$

Interpretation of the terms

In the objective function (3.1), the first term corresponds to the quadratic approximation error of X by $V \cdot \alpha$. The second term is a smoothness regularization equally imposed on each column

of $V \cdot \alpha$ and could also be expressed in an alternate way as:

$$\|\Delta^{1/2}\alpha\|_F^2 = \sum_{i=1}^N \lambda_i \|\alpha_{i,:}\|_2^2,$$

where $\alpha_{i,:}$ is the i -th row of the matrix α . From its definition, we can see that it tends to be low when high values of $\{\lambda_i\}_{i=1}^N$ are associated with rows of α with low ℓ_2 -norm. This corroborates the idea that the $\{\lambda_i\}_{i=1}^N$ can be interpreted as frequencies and the elements of α as Fourier coefficients.

Finally, $\mu_2 \cdot \|\alpha\|_S$, is a sparsity regularization. Authors propose to either use the $\ell_{2,1}$ (the sum of the ℓ_2 -norm of each row of α) or $\ell_{2,0}$ (the number of rows with ℓ_2 -norm different than 0) that induces a row-sparse solution $\hat{\alpha}$. Remark on the choice of $\|\cdot\|_S$. In the context of GSP, it is natural to assume that the graph signals are bandlimited, all at the same dimensions. This property is enforced by $\|\cdot\|_S$ and has two main advantages: it is a key assumption for sampling over a graph and this particular structure is better for inferring graphs with clusters. Therefore, in this article, the use of the classical ℓ_0 -norm and the ℓ_1 -norm has not been investigated since they would impose sparsity at every dimension of the matrix α ‘independently’, which would consequently break the bandlimitedness assumption.

The hyperparameters, $\mu_1, \mu_2 > 0$ are controlling respectively the smoothness of the approximated signals and the sparsity of α . Finally, the first three constraints (1a), (1b), (1c) enforce $V\Delta V^\top$ to be a Laplacian matrix of an undirected graph. More specifically, by definition, $L = D - W$ with $W \in \mathbb{R}_+^{N \times N}$, thus we necessary have $\forall k \neq \ell, L_{k,\ell} = (V\Delta V^\top)_{k,\ell} \leq 0$ (constraint (1b)). Furthermore, as $V\Delta V^\top$ is the eigendecomposition of the Laplacian matrix of an undirected graph, $V^\top V = I_N, v_1 = \frac{1}{\sqrt{N}}\mathbf{1}_N$ and $\lambda_1 = 0 \leq \lambda_2 \leq \dots \leq \lambda_N$ (constraints (1a) and (1c)). The last constraint (1d) was proposed in Dong et al. (2016) as to impose structure in the learned graph while avoiding that the trivial solution $\hat{\Delta} = 0$.

The objective function (3.1) is not jointly convex but when $\|\cdot\|_S$ is taken to be the $\ell_{2,1}$ norm, it is convex with respect to each of the block-variables α, Δ , or V , taken independently.

3.2.5 Problems in the Earlier Proposed Graph Learning Techniques

In the formulations discussed earlier, the focus was primarily on band-limited signals, where only specific frequency components in the spectral domain contribute to the smoothness of graph signals. This approach assumes that the signals are generated solely from a diffusion process, which would indeed justify such a restriction in theory.

However, this assumption does not hold universally in practical scenarios. Many real-world phenomena do not adhere strictly to diffusion processes. Therefore, relying solely on band-limited assumptions may oversimplify the complexity of signal dynamics observed in various applications.

While band-limited models provide a structured framework for understanding signal smoothness, they may overlook important nuances and variations present in real data. In practice, signals often exhibit more complex behaviors and dependencies that cannot be fully captured by a band-limited perspective alone. Thus, there is a need to explore more flexible and adaptive approaches that can accommodate a wider range of signal characteristics and ensure robustness across diverse applications.

3.3 Proposed Formulation for Graph Learning

A New outlook on this problem is needed, as at the onset of Research in this field, assumptions made in the earlier SOTA formulations seemed reasonable, but surely they weren't pragmatic!

3.3.1 Our Proposed Graph Learning Formulation

$$\min_{\alpha, \Delta, V} \|X - V \cdot \alpha\|_F^2 + \mu_1 \cdot \|\alpha\|_{1,1} + \mu_2 \cdot \|V \Delta V^\top\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^\top - \frac{1}{\sigma^2} \cdot I\|_F^2$$

$$\text{s.t.} \quad \begin{cases} V^\top V = I_N, & \text{(a)} \\ \frac{1}{\sqrt{N}} \mathbb{1}'_N \cdot V = [100 \cdots 0], & \text{(b)} \\ \Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \succeq 0, & \text{(c)} \\ V \Delta V^\top \cdot \mathbb{1}_N = \mathbb{0}_N, & \text{(d)} \\ \text{diag}(V \Delta V^\top) \geq 0, & \text{(e)} \\ V \Delta V^\top - \text{diag}(\text{diag}(V \Delta V^\top)) \leq 0, & \text{(f)} \\ \text{tr}(\Delta) = \text{tr}(\hat{\Theta}) - N \cdot \frac{1}{\sigma^2}, & \text{(g)} \end{cases} \quad (3.2)$$

Explanation of the Terms:

Our objective function balances three crucial components:

1. **Reconstruction Error Minimization:** Minimize the error between the observed data matrix X and its approximation using spectral coefficients α and basis vectors V .
2. **Promotion of Sparsity in α :** Encourage a parsimonious representation of spectral signals by promoting sparsity in α .
3. **Fidelity to Target Graph Structure $\hat{\Theta}$:** Enforce fidelity to a target graph structure $\hat{\Theta}$ using a quadratic penalty term. This ensures alignment with observed data while considering noise factors.

The constraints imposed ensure the integrity and appropriateness of the learned graph structures:

- (a) V is orthogonal ($V^\top V = I_N$), preserving the geometric integrity of the graph.
- (b) V aligns the first row to a specific pattern ($\frac{1}{\sqrt{N}} \mathbb{1}'_N \cdot V = [100 \cdots 0]$), reflecting practical constraints on graph structures.
- (c) Δ is a diagonal matrix with non-negative eigenvalues ($\Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \succeq 0$), capturing the intrinsic sparsity of graph Laplacians.
- (d) $V \Delta V^\top$ annihilates global signals ($V \Delta V^\top \cdot \mathbb{1}_N = \mathbb{0}_N$), ensuring local connectivity.
- (e) The diagonal of $V \Delta V^\top$ enforces non-negativity ($\text{diag}(V \Delta V^\top) \geq 0$), crucial for meaningful interpretations.
- (f) The non-diagonal elements of $V \Delta V^\top$ are non-positive ($V \Delta V^\top - \text{diag}(\text{diag}(V \Delta V^\top)) \leq 0$), maintaining consistency with graph theoretic norms.
- (g) The trace condition on Δ ensures energy conservation ($\text{tr}(\Delta) = \text{tr}(\hat{\Theta}) - N \cdot \frac{1}{\sigma^2}$), aligning the learned graph with empirical data observations.

This formulation integrates these principles to provide a comprehensive approach for learning graph structures that are both sparse and adaptive to spectral variations in the data. By emphasizing these aspects, it enhances the robustness and interpretability of graph learning methodologies in diverse practical applications.

3.3.2 Solving Approach

The formulation is not jointly convex hence to solve this problem we alternate between the three variables, minimizing one while keeping the others fixed as shown below.

Algorithm 1 Our Graph Learning Algorithm

Input: $X \in \mathbb{R}^{N \times T}$, $\sigma_{oracle} \triangleright$ Signals i.e. X : T Observations across N Entities in a Population

Output: α^*, Δ^*, V^*

Initialization:

$[V, \Delta, U] \leftarrow \text{SVD}(X)$

$v_1 \leftarrow \frac{1}{\sqrt{N}} \mathbf{1}_N$

$\triangleright$ Make first column of V as $\frac{1}{\sqrt{N}} \mathbf{1}_N$

$V \leftarrow \text{QR}(V)$

$\mu \leftarrow \text{mean}(X)$

$\triangleright \mu \in \mathbb{R}^{N \times 1}$: Mean of observations over each Node

$\hat{\Theta} \leftarrow \left(\frac{(X - \mu) \cdot (X - \mu)^\top}{T - 1} \right)^{-1}$

$\triangleright \hat{\Theta} \in \mathbb{R}^{N \times N}$: Precision Matrix or Inverse Covariance Matrix

$\sigma = \sigma_{oracle}$

while not convergence **do**

$\triangleright$ α -step: Learning latent representation of the Signals in the Fourier Space.

$$\min_{\alpha} \|X - V \cdot \alpha\|_F^2 + \mu_1 \cdot \|\alpha\|_{1,1}$$

$\triangleright$ Δ -step: Learning Eigen Values of the Graph Laplacian L .

$$\min_{\Delta} \mu_2 \cdot \|V \Delta V^\top\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^\top - \frac{1}{\sigma^2} \cdot I\|_F^2$$

$$\text{s.t.} \quad \begin{cases} \Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \succeq 0, & \text{(a)} \\ V \Delta V' \cdot \mathbf{1}_N = \mathbf{0}_N, & \text{(b)} \\ \text{diag}(V \Delta V') \geq 0, & \text{(c)} \\ V \Delta V' - \text{diag}(\text{diag}(V \Delta V')) \leq 0, & \text{(d)} \\ \text{tr}(\Delta) = \text{tr}(\hat{\Theta}) - N \cdot \frac{1}{\sigma^2}, & \text{(e)} \end{cases}$$

$\triangleright$ V -step: Learning Eigen Vectors of the Graph Laplacian L .

$$\min_V \|X - V \cdot \alpha\|_F^2 + \mu_2 \cdot \|V \Delta V^\top\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^\top - \frac{1}{\sigma^2} \cdot I\|_F^2$$

$$\text{s.t.} \quad \begin{cases} V^\top \cdot V = I_N, & \text{(a)} \\ \frac{1}{\sqrt{N}} \mathbf{1}_N' \cdot V = [1 \ 0 \ \dots \ 0], & \text{(b)} \\ V \Delta V' \cdot \mathbf{1}_N = \mathbf{0}_N, & \text{(c)} \\ \text{diag}(V \Delta V') \geq 0, & \text{(d)} \\ V \Delta V' - \text{diag}(\text{diag}(V \Delta V')) \leq 0, & \text{(e)} \end{cases}$$

end while

In the above 3-step alternate minimization procedure, we learn the representation of the signals in the Fourier space with basis as the Eigenvectors of the graph Laplacian corresponding to the underlying Graphical relation within the Entities in the population. Also, the basis of the Fourier space i.e. the Eigenvectors and thus the corresponding Eigenvalues of the Graph Laplacian are being learnt.

3.3.3 Proposed Sparsity-Based Graph Signal Generation Model

Given a Laplacian matrix $L = V\Delta V^\top$, we propose the following **Factor Analysis Framework** to model a graph signal $x \in \mathbb{R}^N$

$$x = V \cdot \alpha + m_y + \varepsilon ,$$

where $m_y \in \mathbb{R}^N$ is the mean of the graph signal x and ε is a Gaussian noise with zero mean and covariance $\sigma^2 I_N$. Here, the latent variable $\alpha = [\alpha_1, \dots, \alpha_N]^\top$ controls x through the eigenvector matrix V of L . The choice of the representation matrix V is particularly adapted since it reflects the topology of the graph and provides a spectral embedding of its vertices. V can be interpreted as a graph Fourier basis, which makes it an intuitive choice for the representation matrix. In a noiseless scenario with $m_y = 0$, α actually corresponds to the **Graph Fourier Transform** of x .

To comply with the spectral sparsity, we now propose a distribution that allows α to admit zero-valued components. To this end, we introduce independent latent Bernoulli variables γ_i with success probability $p_i \in [0, 1]$. Knowing $\gamma_1, \dots, \gamma_N$, the conditional distribution for α is:

$$\alpha|\gamma \sim \mathcal{N}(0, \tilde{\Lambda}^\dagger)$$

where $\tilde{\Lambda}^\dagger$ is the Moore-Penrose pseudo-inverse of the diagonal matrix containing the values $\{\lambda_i \mathbb{1}\{\gamma_i = 1\}\}_{i=1}^N$. In this model, γ_i controls the sparsity of the i -th element of α . Indeed, if $\gamma_i = 0$, then $\alpha_i = 0$ almost surely. In the other hand, if $\gamma_i = 1$ then α_i follows a Gaussian distribution with zero-mean and variance equal to $1/\lambda_i$. This is adapted to the smoothness hypothesis as for a high value of λ_i (high frequency), the distribution of α_i concentrates more around 0, leading to the small value of $\lambda_i \alpha_i^2$.

We introduce sparsity by choosing the probabilities at random, thereby discouraging bandlimitedness and encouraging low-frequency and high-frequency components of the basis to take part in the signal generation process because a realistic evolution process of a signal may not have resulted from a pure diffusion process.

Note that, since $\lambda_1 = 0$, α_1 follows a centered degenerate Gaussian, i.e. α_1 is equal to 0 almost surely. Furthermore, if $p_i = 1$ for all i , our model reduces to the one proposed by Dong et al. (2016), which was only focused on the smoothness assumption.

Summarization: The following equations summarize the **Prior** and important **conditional distributions** of our model:

$$p(\alpha_i|\gamma_i, \lambda_i) \propto \exp(-\lambda_i \cdot \alpha_i^2) \mathbb{1}\{\gamma_i = 1\} + \mathbb{1}\{\alpha_i = 0, \gamma_i = 0\} ,$$

$$p(x|\alpha, V) \propto \exp\left(-\frac{1}{\sigma^2} \|x - V \cdot \alpha - m_y\|_2^2\right) ,$$

$$p(\gamma_i) \propto p_i^{\gamma_i} (1 - p_i)^{1-\gamma_i} .$$

For simplicity, in the following, we consider that $m_y = 0$ and $p_1 = 0$.

3.4 Results

3.4.1 Standard Graphs for Performing Synthetic Experiments

The Different Types of Standard Graphs used are:

1. A **Random Geometric** graph is constructed using a 2-D uniform distribution for the node coordinates and a truncated Gaussian kernel with a width of 0.5 for the edges. Any edge weights smaller than 0.75 are set to 0. [N = 20 Nodes]
2. An **Erdős-Rényi** graph model is constructed with an edge creation probability of 0.2. [N = 20 Nodes]
3. The **Barabási Albert** graph model is used to generate a scale-free random graph by adding one new vertex to the graph at each time, connecting it to one existing vertex in the graph, with the probability of the new vertex being linked to a given existing vertex in the graph being proportional to the ratio of the degree of that existing vertex to the sum of degrees of all the existing vertices. [N = 20 Nodes]
4. A **Forest Fire** graph is generated with *forest fire forward/backward burning* probability equal to 0.3. [N = 20 Nodes]
5. A **Stochastic Kronecker** graph model is constructed using an initiator graph of size 3×3 with weights [0.75, 0.65, 0.55]. Since we consider undirected graphs in this paper, the directed graphs returned by the above graph model are symmetrized. [N = 3^3 Nodes]
6. The **Zachary's Karate Club** Graph is constructed with a node attribute 'club' indicating the club to which the represented member belongs ('Mr. Hi' or 'Officer'), and edges with weights based on the number of interactions between the incident node members. [N = 34 Nodes]
7. The **Watts Strogatz** graph model is used to generate a connected small-world graph. First, create a ring over n nodes. Then each node in the ring is joined to its k nearest neighbors (or $k - 1$ neighbors if k is odd). Then shortcuts are created by replacing some edges as follows: for each edge, (u, v) in the underlying " n -ring with k nearest neighbors" with probability p replace it with a new edge (u, w) with uniformly random choice of existing node w . The entire process is repeated until a connected graph results. [n = 20, k = 4, p = 0.1]
8. The **Dorogovtsev, Goltsev & Mendes** graph model begins with a single edge connecting two vertices at $t = -1$. At each time step, a new vertex is added to every edge of the graph, attaching to both end vertices of the edge. By $t = 0$, a triangle of edges connects a triple of vertices, and by $t = 1$, the graph consists of 6 vertices connected by 9 edges, and so on. The total number of vertices at "time" t is $N_t = 3(3^t + 1)/2$, and the total number of edges is $L_t = 3^{t+1}$, resulting in an average degree of $\bar{k}_t = 2L_t/N_t = 4/(1 + 3^{-t})$. [N = 15 Nodes]
9. The **Stochastic Block Model** is utilized to construct a sparse but connected graph with C clusters and about $k\%$ of the total edges possible in an undirected graph i.e. $\frac{N \cdot (N-1)}{2}$. Initially, a random tree graph with the same number of nodes is generated. Subsequently, the nodes are divided into C clusters, and τ edges are added within each cluster. Finally, excess edges beyond the maximum allowed percentage of edges are removed while ensuring connectivity. In the event of disconnectivity, a pair of nodes from different disconnected

components are randomly selected, and an edge is added between them until the graph becomes connected. The last 2 steps are repeated until graph connectivity and the maximum edge limit criterion are satisfied. [$N = 20$ Nodes, $k = 25\%$ Edges, $\tau = 3$]

3.4.2 Experiments on Standard Graphs with Graph Signals defined as per Our Proposed Sparse Graph Signal Generation Model

In our study, a standard graph is constructed where signals are defined according to our proposed signal generation model. The model synthesizes data from various sources to form a comprehensive view of the signal characteristics across the graph nodes.

Results Overview The results of our analysis are summarized on the following pages. Key observations and insights are highlighted below.

Commentary on Results Our findings indicate significant advancements compared to state-of-the-art (SOTA) techniques in the field.

F-Scores: We consistently outperform nearly all existing techniques in terms of F-scores, showcasing the effectiveness of our approach in accurately capturing relationships and patterns within the data. Relative RMSEs in Reconstructed Matrices: Even when competing against other techniques, our method demonstrates superior performance in terms of relative root mean square errors (RMSEs) in the reconstructed matrices. This underscores our model’s ability to reconstruct data with higher fidelity and accuracy. *All Results for Synthetic Signals collated at the END.*

3.4.3 Baseline Comparison: Assessing Performance with Smooth Signals

To further evaluate our technique’s performance with respect to smooth signals, we conducted baseline tests across all methodologies. Notably, our results show comparable performance across these tests, reinforcing the robustness of our methodology.

Conclusion These findings affirm that our proposed methodology effectively addresses the complexity of signal generation and representation in graph-based models. By emphasizing sparsity and leveraging advanced signal processing techniques, our approach not only excels in performance metrics like F-scores but also maintains competitiveness in scenarios requiring smooth signal handling.

This comprehensive evaluation underscores the capability of our methodology to meet the diverse challenges posed by real-world data applications.

This shows the Proposed Method is indeed able to do justice to ‘Sparsity’.

3.4.4 Experiments on Real Data

1. US County Temperature Dataset:

The US temperature dataset is a comprehensive collection of temperature records from 45 counties, each with 5,844 time steps. The primary goal is to construct a graph that connects counties with similar temperature evolution over the years.

Challenges in Ground Truth Graph Construction

Creating a ground truth graph for this dataset based solely on altitude presents significant challenges. According to Ortega’s research, altitude is not the sole factor influencing similar temperature trends. His study indicates that other latent factors play a critical role in the evolution

of temperature patterns. This makes it difficult to rely only on altitude to establish connections between counties.

Demonstrating Model Capabilities

This dataset serves as an excellent example to showcase our model’s capability to balance ‘sparsity without losing smoothness’. To illustrate, consider the graph learned by our algorithm, which includes an edge connecting California and Florida. At first glance, one might argue that these two locations do not have direct heat transfer interactions, suggesting that this edge is a ‘spurious link’.

Insights from Ortega’s Method

Ortega’s method, based on diffusion processes and the heat kernel, predominantly focuses on smoothness—spreading from one node to its adjacent nodes. As a result, it fails to capture global relationships like the one between California and Florida. Ortega’s graph does not show this link, supporting his conclusion that diffusion-based algorithms cannot identify such global patterns.

Comparison with Dong’s Method

Dong’s method, which emphasizes global smoothness, successfully captures the link between California and Florida. However, it falls short in identifying other known regional similarities. This demonstrates that while global smoothness is important, it is not sufficient to comprehensively understand all temperature relationships in the dataset.

Advantages of Our Algorithm

Our algorithm excels by integrating both local and global relationships. It captures the link between California and Florida, as well as other regional similarities, by considering sparsity in both geometric and spectral spaces. This dual consideration allows our model to identify specific edges while maintaining the general heat transfer trends.

Implications of Our Findings

The ability of our algorithm to capture both sparsity and smoothness highlights its robustness and versatility. By acknowledging that smoothness is a special case of sparsity, our model can generalize better across different datasets and scenarios. This formulation ensures that we do not miss out on critical connections, whether they are local or global in nature.

Conclusion

In summary, our formulation on the US temperature dataset effectively demonstrates that smoothness can be viewed as a special case of sparsity. This unique approach enables our algorithm to capture both specific and general relationships within the data, showcasing its superior performance and generalizability compared to other methods.

US temperature Variation Dataset over 45 Counties with each having 5844 temperature readings

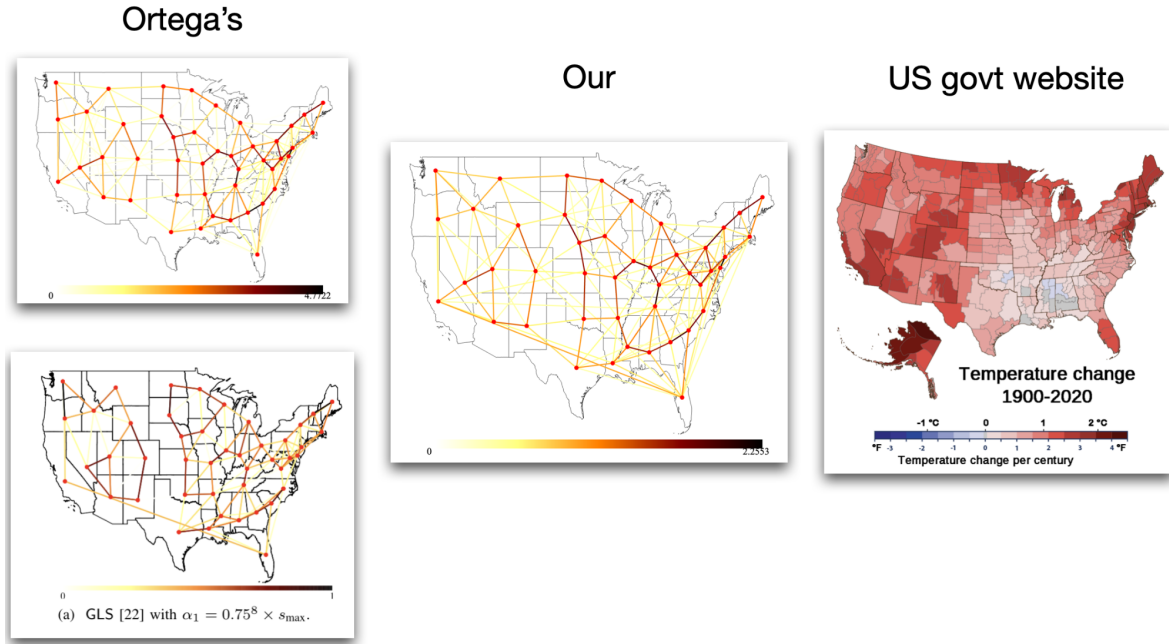


Figure 3.11: Visual Comparison

2. Animal Clustering Dataset:

The Animal dataset comprises 33 animals, each characterized by responses to 102 binary questions (e.g., ‘Lives in water?’ or ‘Can fly?’). For each animal, the answer to each question is recorded as either 0 or 1, resulting in a feature set of 102 binary signals per animal. The primary goal was to cluster similar animals together based on these responses.

Characteristics of the Learned Graph

The graph learned by our algorithm for this dataset is disconnected, reflecting distinct clusters of animals. This clustering appears to be quite reasonable, aligning well with intuitive groupings of the animals based on their features.

Key Attributes of the Animal Dataset

Non-Gaussian Nature: This dataset is non-Gaussian, showcasing our model’s ability to perform well in model-mismatch settings where traditional Gaussian assumptions do not hold. **Binary Dataset with Limited Signals:** With only 102 binary signals, this dataset demonstrates our model’s effectiveness in learning with a relatively small number of features. **Disconnected Components:** The learned graph exhibits our model’s capability to form a graph with disconnected components, a feature not readily achieved by other methods.

Comparison with Other Methods

Other algorithms applied to this dataset require a pre-assigned parameter k , representing the number of clusters. For instance, if $k=5$, those algorithms learn a graph with exactly five components. In contrast, our algorithm does not necessitate specifying k beforehand. Instead, it controls the sparsity of the learned graph both geometrically and in spectral space, offering a more general and robust approach.

Implications of Our Findings

Our algorithm’s ability to learn a graph without needing a pre-determined number of clusters highlights its flexibility and adaptability. By focusing on sparsity, our method is more robust and generalizable, capable of handling various data types and structures, including those with binary

signals and non-Gaussian distributions.

Conclusion

In summary, the Animal dataset exemplifies our model’s strengths in handling non-Gaussian data, learning from a limited number of binary signals, and forming graphs with disconnected components. The learned graph reflects reasonable clustering of animals, underscoring the robustness and versatility of our algorithm compared to other methods that require pre-assigned clustering parameters.

ANIMAL DATASET - 33 Animals with each having 102 Features

clustering_stats (Done on Basis of Pair-Wise Presence)

Author	Precision	Recall/Sensitivity	F-Score	Specificity
Our	0.959	0.887	0.922	0.996
Dong	0.627	0.792	0.700	0.947
Humbert	0.880	0.830	0.854	0.987
Ortega	0.578	0.491	0.531	0.960
Palomar	0.773	0.642	0.701	0.979
Nie	0.915	0.811	0.860	0.992
SGL	0.656	0.792	0.718	0.954

Figure 3.12: Quantitative Comparison

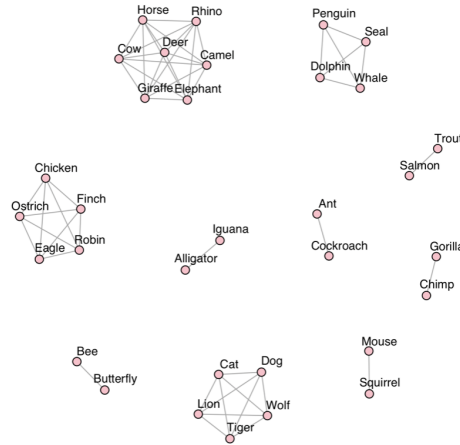


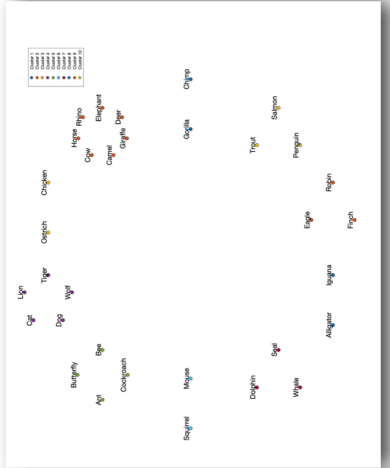
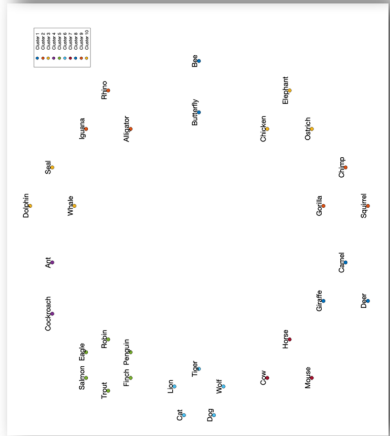
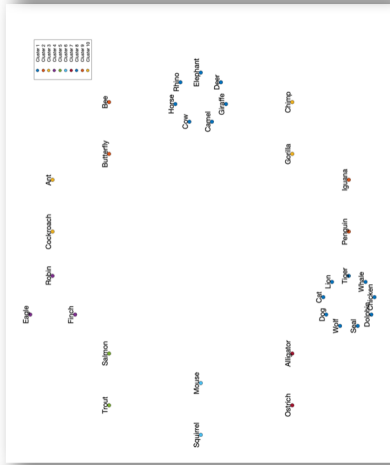
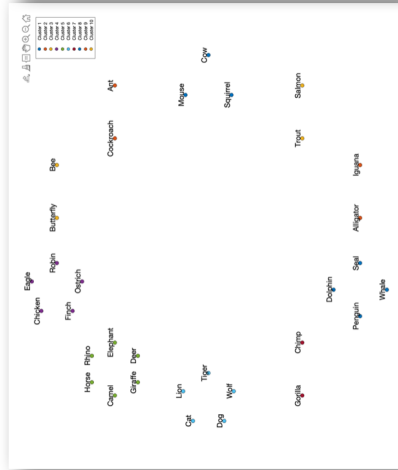
Figure 3.13: Standard 10 Cluster Animal Dataset

Our 9

Dong 81

Ortega

Humbert 76



Palomar 6

Nie et al k=10 1

SGL k=10 39

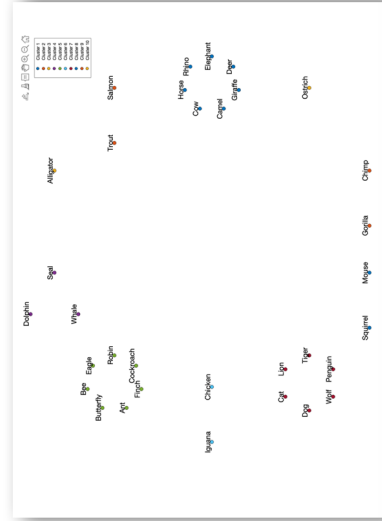
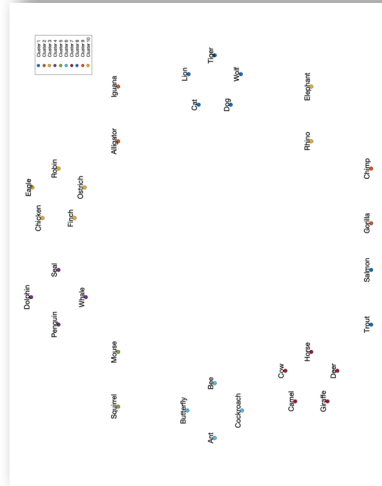
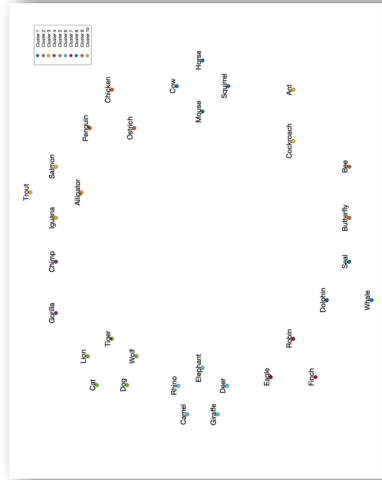


Figure 3.14: Visual Comparison

3. US Senate Roll Call - Opinion Based Republican-Democratic US County Classification :

The US Senate Rollcall Records dataset consists of the opinions of senators from the 110th US Congress (2007-2009) on 657 roll calls. Each state's senators' opinions were categorized into four groups and then merged using a specific protocol, resulting in a dataset with 50 nodes (one per state) and 657 signals (one per roll call).

Objective and Methodology

The primary goal was to cluster the states into two ideological groups based on their senators' opinions. To achieve this:

Our Graph Learning Formulation was applied to the data matrix. Spectral Clustering was used on the learned graph to create two clusters.

Progress and Results

The clustering results so far appear impressive. The process involves a brute force search based on cross-validation error to identify the optimal set of hyperparameters. While this search is ongoing, the current results indicate a promising clustering that is expected to improve with further refinement of the hyperparameters.

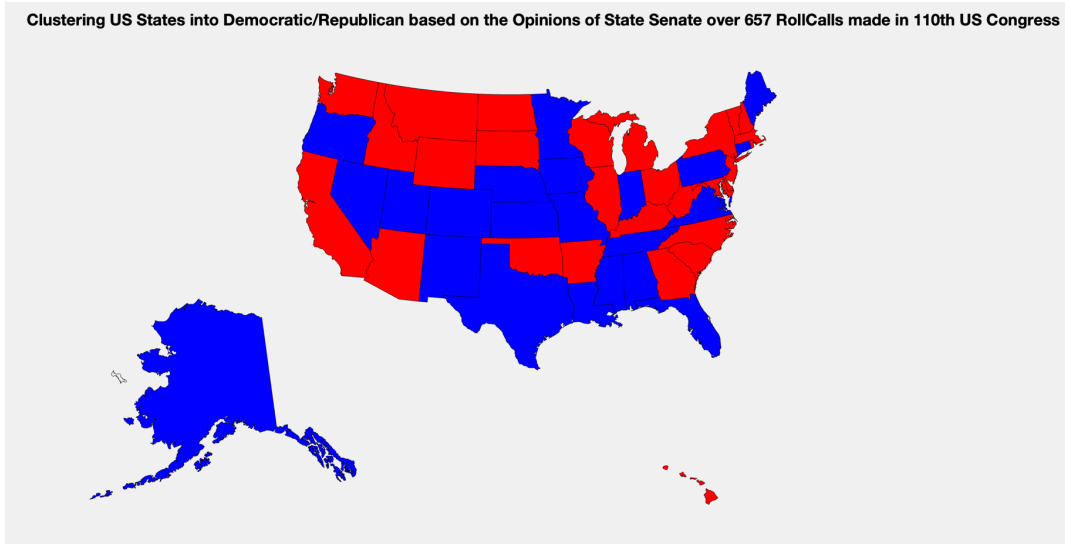


Figure 3.15: Red-Blue States in the US

Chapter 4

Hyper-parameter Selection Strategy

4.1 Inconsistency in Model Selection for Other State-of-the-Art Techniques

The lack of consistency in selecting the appropriate threshold post-learning in graph models poses a significant challenge. To address this issue, we propose a principled approach that allows for the optimal selection of parameters in a purely data-driven manner. By leveraging observed data and rigorous statistical methods, our approach ensures that thresholds and parameters are chosen systematically, enhancing the reliability and interpretability of the learned graph structures. This methodological advancement not only improves the accuracy of graph-based analyses but also provides a standardized framework applicable across various graph learning techniques, thereby promoting consistency and reproducibility in the field.

4.2 Proposed Principled Approach for Hyper-parameter Estimation

Proposed Algorithm on the Next Page.

Note:

1. Find the combination of $\mu_1, \mu_2, \mu_3, \tau$ that yields the least cross-validation error.
2. Deem that combination as the best hyperparameter choice and thus re-estimate the graph using our algorithm (3.2) overall signals X .

4.3 Applicability to Earlier SOTA Work

The proposed strategy is inherently versatile and data-driven, leveraging observed signals to inform its methodology. It operates by first learning from one-half of a split dataset and then employing this learned model to reconstruct the other half. This task is fundamental in the realm of Graph Signal Processing (GSP), where the goal is to infer properties or patterns across graph-structured data. Thus, this principled approach holds applicability across various formulations within the field of GSP. By grounding its decisions in observed data, this method ensures adaptability and robustness in handling diverse datasets and scenarios encountered in practical applications of graph learning.

Algorithm 2 Algorithm for Cross-Validation Based Hyper-parameter tuning

Input: $X \in \mathbb{R}^{N \times T}$, $\sigma_{oracle}^R, \sigma_{oracle}, \kappa$ ▷ Signals i.e. X, κ for Train-Validation Split, τ_r : Thresholds
Output: $\mu_1^*, \mu_2^*, \mu_3^*, \tau$
Initialization for X_R and y :
 $X_R \leftarrow \kappa\%$ of X ▷ $\kappa\%$ of Signals in chosen for Training i.e. Reconstruction Set
 $y \leftarrow X - X_R$ ▷ y Represents the Validation Set.
 $[V_R, \Delta, U_R] \leftarrow \text{SVD}(X_R)$
 $v_1 \leftarrow \frac{1}{\sqrt{N}} \mathbf{1}_N$ ▷ Make first column of V as $\frac{1}{\sqrt{N}} \mathbf{1}_N$
 $V_R \leftarrow \text{QR}(V_R)$
 $\mu_R \leftarrow \text{mean}(X_R)$ ▷ $\mu_R \in \mathbb{R}^{N \times 1}$: Mean of observations over each Node
 $\hat{\Theta}_R \leftarrow \left(\frac{(X_R - \mu_R) \cdot (X_R - \mu_R)^\top}{T_R - 1} \right)^{-1}$ ▷ $\hat{\Theta} \in \mathbb{R}^{N \times N}$: Precision Matrix or Inverse Covariance Matrix
 $\sigma^R = \sigma_{oracle}^R$
Initialization for X :
 $[V, \Delta, U] \leftarrow \text{SVD}(X)$
 $v_1 \leftarrow \frac{1}{\sqrt{N}} \mathbf{1}_N$ ▷ Make first column of V as $\frac{1}{\sqrt{N}} \mathbf{1}_N$
 $V \leftarrow \text{QR}(V)$
 $\mu \leftarrow \text{mean}(X)$ ▷ $\mu \in \mathbb{R}^{N \times 1}$: Mean of observations over each Node
 $\hat{\Theta} \leftarrow \left(\frac{(X - \mu) \cdot (X - \mu)^\top}{T - 1} \right)^{-1}$ ▷ $\hat{\Theta} \in \mathbb{R}^{N \times N}$: Precision Matrix or Inverse Covariance Matrix
 $\sigma = \sigma_{oracle}$
for all μ_1, μ_2, μ_3 do
▷ Learning α_R, Δ_R, V_R over the Reconstruction Set.

$$\min_{\alpha_R, \Delta_R, V_R} \|X_R - V_R \cdot \alpha_R\|_F^2 + \mu_1 \cdot \|\alpha_R\|_{1,1} + \mu_2 \cdot \|V_R \Delta_R V_R^\top\|_{1,1} + \mu_3 \cdot \|\hat{\Theta}_R - V_R \Delta_R V_R^\top - \frac{1}{\sigma_R^2} \cdot I\|_F^2$$

$$\text{s.t.} \quad \begin{cases} V_R^\top \cdot V_R = I_N, & \text{(a)} \\ \frac{1}{\sqrt{N}} \mathbf{1}_N' \cdot V_R = [100 \dots 0], & \text{(b)} \\ \Delta_R = \text{diag}(0, \lambda_2, \dots, \lambda_N) \succeq 0, & \text{(c)} \\ V_R \Delta_R V_R' \cdot \mathbf{1}_N = \mathbf{0}_N, & \text{(d)} \\ \text{diag}(V_R \Delta_R V_R') \geq 0, & \text{(e)} \\ V_R \Delta_R V_R' - \text{diag}(\text{diag}(V_R \Delta_R V_R')) \leq 0, & \text{(f)} \end{cases}$$
▷ Estimate Matrices L_R, W_R using learned V_R, Δ_R .

$$L_R = V_R \cdot \Delta_R \cdot V_R^\top$$

$$D_R = \text{diag}(\text{diag}(L_R))$$

$$W_R = D_R - L_R$$
▷ Identifying the best τ that minimises the validation error ve_τ over validation set y .
for all τ do

$$W_\tau = W_R > \tau$$

$$(D_\tau)_{i,i} = \sum_j (W_\tau)_{i,j}$$

$$L_\tau = D_\tau - W_\tau$$

$$[V_\tau, \Delta_\tau] = \text{eig}(L_\tau)$$

$$\alpha_\tau = \min_{\alpha_\tau} \|y - V_\tau \cdot \alpha_\tau\|_F^2 + \mu_1 \cdot \|\alpha_\tau\|_{1,1}$$

$$ve_\tau = \|y - V_\tau \cdot \alpha_\tau\|_F^2$$
end for
▷ Store for the Combination μ_1, μ_2, μ_3 that τ which gave the least ve_τ .
end for

Figure 4.1: Cross Validation Based Hyperparameter Tuning Strategy

Chapter 5

Streaming Algorithm

5.1 Introduction to Streaming Algorithm

Enhancing algorithm convergence time is pivotal because if not then performing principled hyper-parameter estimation poses a challenge. We devised a novel approach leveraging the existing graph's value, with minimal adjustments for tasks like node insertion, update, or deletion, facilitating efficient learning of the new graph.

We conduct comprehensive testing of our Streaming algorithm using both Synthetic and Real Datasets to thoroughly assess the efficacy and robustness of our proposed model. Through this extensive evaluation process, we aim to quantify the performance and validate the applicability of our algorithm across various scenarios.

In a nutshell, the idea is simply to update the graph based on the addition of a new node.

5.2 Streaming Algorithm for Dong et. al.'s Formulation

Here we first test the methodology on a standard SOTA technique in GSP i.e. on Dongs implementation of graph learning problem. Given below is the Standard Graph Learning Formulation implemented by Dong. The image after that shows Graph Learning with Streaming modified by us.

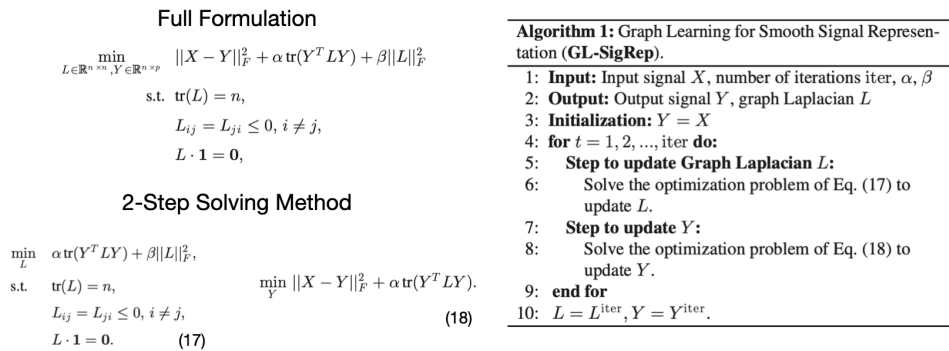


Figure 5.1: Dong's Formulation

Algorithm 1 Streaming Formulation for Dong et. al. Algorithm

Input: L^o Represents the $N - 1$ Dimensional Graph Laplacian

Initialisation:

$$y^s = Y(1 : N - 1, 1 : N - 1)$$

$$y_{lc} = Y(:, N)$$

Output: L Represents the N Dimensional Graph Laplacian

while not convergence **do**

▷ ℓ -step: Learning the Last Row Vector of the New Laplacian

$$\begin{aligned} & \min_{\ell} \alpha \cdot [\text{tr}(L^o \cdot y^s) + (\ell \cdot y_{lc}) \cdot 2 - \ell(N) \cdot y(N, N) - \ell(1 : N - 1) \cdot \text{diag}(y^s)] \\ & + \beta \cdot [\|\ell\|_F^2 + \|\ell(1 : N - 1)\|_F^2 + \|L^o - \text{diag}(\ell(1 : N - 1))\|_F^2] \\ & \text{s.t.} \quad \begin{cases} \ell(1 : N - 1) \leq 0, & \text{(a)} \\ \text{sum}(\ell) = 0, & \text{(b)} \\ \ell(N, N) = 1 + \text{sum}(\ell(1 : N - 1)), & \text{(c)} \end{cases} \end{aligned} \tag{5.1}$$

▷ Update L using ℓ :

$$\begin{aligned} L(1 : N - 1, 1 : N - 1) &= L^o - \text{diag}(\ell(1 : N - 1)) \\ L(N, :) &= \ell \\ L(:, N) &= \ell' \end{aligned}$$

▷ Y -step: Learning equivalent representation of the Signals.

$$\min_Y \|X - Y\|_F^2 + \lambda \cdot \text{tr}(Y'LY) \tag{5.2}$$

end while

5.2.1 Results on Streaming with Dongs Implementation

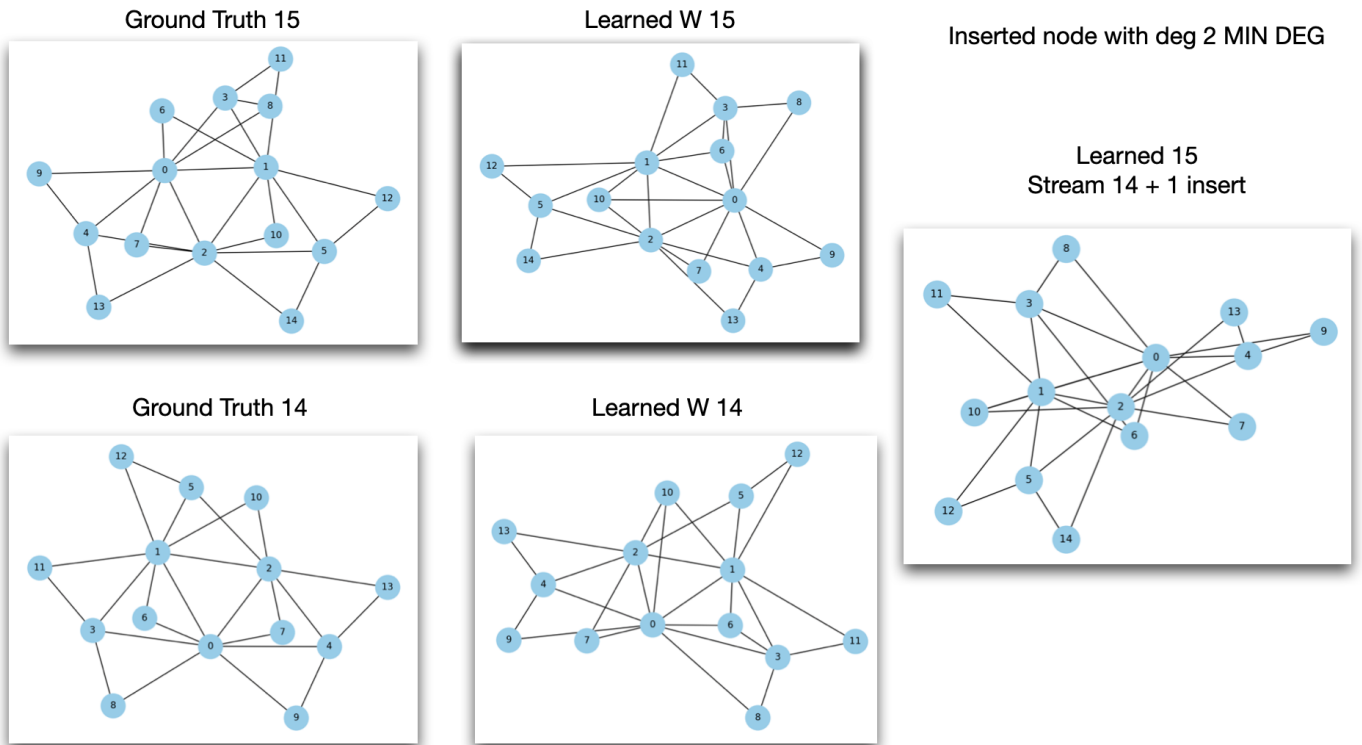


Figure 5.2: Graph 1

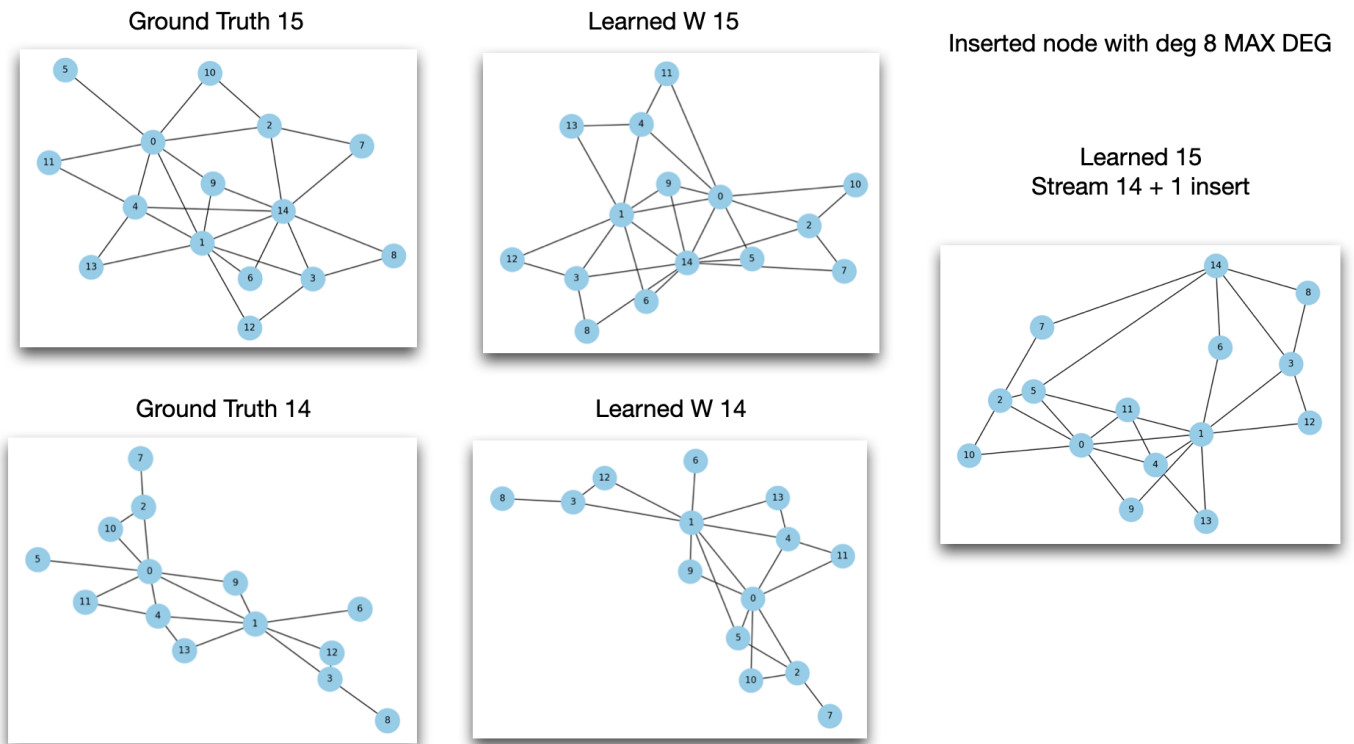


Figure 5.3: Graph 2

5.3 Proposed Streaming Algorithm for Our Formulation

In this thesis, we propose a novel Graph Learning-Based Streaming formulation designed to extend beyond the constraints of handling only smooth signals, as traditionally addressed in the literature. Our formulation not only accommodates smooth signals, as discussed by Dongs et al., but also effectively caters to sparse signals, which are prevalent in many practical scenarios. By leveraging graph structures and algorithms tailored for both smoothness and sparsity, our approach aims to enhance the robustness and applicability of graph-based learning methods in real-world applications where signals exhibit varying degrees of regularity and concentration. This dual capability is pivotal in advancing the state-of-the-art in graph learning, offering new insights and potential solutions to challenging problems in signal processing and data analysis.

$$\begin{aligned}
& \min_{\alpha, \Delta, V} \|X - V \cdot \alpha\|_F^2 + \mu_1 \cdot \|\alpha\|_{1,1} + \mu_2 \cdot \|V \Delta V^\top\|_{1,1} + \mu_3 \cdot \|\hat{\Theta} - V \Delta V^\top - \frac{1}{\sigma^2} \cdot I\|_F^2 \\
& \text{s.t.} \quad \begin{cases} V^\top V = I_N, & \text{(a)} \\ \frac{1}{\sqrt{N}} \mathbf{1}'_N \cdot V = [100 \cdots 0], & \text{(b)} \\ \Delta = \text{diag}(0, \lambda_2, \dots, \lambda_N) \succeq 0, & \text{(c)} \\ V \Delta V' \cdot \mathbf{1}_N = \mathbf{0}_N, & \text{(d)} \\ \text{diag}(V \Delta V') \geq 0, & \text{(e)} \\ V \Delta V' - \text{diag}(\text{diag}(V \Delta V')) \leq 0, & \text{(f)} \\ \text{tr}(\Delta) = \text{tr}(\hat{\Theta}) - N \cdot \frac{1}{\sigma^2}, & \text{(g)} \end{cases}
\end{aligned}$$

Figure 5.4: Our Formulation

In the realm of signal processing and graph theory, the proposed streaming algorithm aims to update the Graph using the older lower dimension graph by taking help of only the new node that got entered in the population set. Below is the proposed algorithm to do this:

$$\begin{aligned}
& \min_{L, Y} \|X - Y\|_F^2 + \mu_1 \cdot \|Y\|_F^2 + \mu_2 \cdot \|L\|_F^2 + \mu_3 \cdot \|\hat{\Theta} - L - \frac{1}{\sigma^2} \cdot I\|_F^2 \\
& \text{s.t.} \quad \begin{cases} L \cdot \mathbf{1}_N = \mathbf{0}_N, & \text{(a)} \\ \text{diag}(L) \geq 0, & \text{(b)} \\ \text{off-diag}(L) \leq 0, & \text{(c)} \\ \text{tr}(L) = \text{tr}(\hat{\Theta}) - N \cdot \frac{1}{\sigma^2}, & \text{(d)} \end{cases}
\end{aligned}$$

This algorithm is formulated as a minimization problem, which is a combination of four key terms. The first term, $\|X - Y\|_F^2$, measures the difference between the input data matrix, X , and the estimated signal matrix, Y . This term ensures that the estimated signals are as close as possible to the original data. The second term, $\mu_1 \cdot \|Y\|_F^2$, is a regularization term on the estimated signals. The third term, $\mu_2 \cdot \|L\|_F^2$, is a regularization term on the graph Laplacian, L , which encourages sparsity and smoothness in the graph structure. The fourth term, $\mu_3 \cdot \|\hat{\Theta} - L - \frac{1}{\sigma^2} \cdot I\|_F^2$, ensures that the graph Laplacian, L , is close to the inverse covariance matrix, $\hat{\Theta}$ which brings the fidelity of the Data, adjusted for the variance, σ^2 .

The optimization problem is subject to several constraints to ensure the validity of the graph Laplacian, L . The first constraint, $L \cdot \mathbf{1}_N = \mathbf{0}_N$, ensures that the sum of each row in the graph

Laplacian is zero, which is a fundamental property of Laplacian matrices. The second constraint, $\text{diag}(L) \geq 0$, ensures that the diagonal elements of the graph Laplacian are non-negative. The third constraint, $\text{off-diag}(L) \leq 0$, ensures that the off-diagonal elements of the graph Laplacian are non-positive, which implies that the weights of the graph edges are non-negative. The fourth constraint, $\text{tr}(L) = \text{tr}(\hat{\Theta}) - N \cdot \frac{1}{\sigma^2}$, ensures that the trace of the graph Laplacian is equal to the trace of the adjusted inverse covariance matrix. This constraint is crucial for maintaining the consistency between the graph structure and the statistical properties of the data. Below are the Algorithmic Implementation Details presented in Step-by-Step Fashion:

Algorithm 2 Streaming Formulation for Our Algorithm

Input: L^o Represents the $N - 1$ Dimensional Graph Laplacian

Output: L Represents the N Dimensional Graph Laplacian

while not convergence **do**

▷ ℓ -step: Learning the $N-1$ Elements of the Last Row of the New Laplacian Matrix

$$\begin{aligned} \min_{\ell} \mu_2 \cdot & \left[\|(L^o - \text{diag}(\ell))\|_F^2 + 2\|\ell\|_F^2 + \text{sum}(\ell)^2 \right] \\ & + \mu_3 \cdot \left[\left\| \hat{\theta}(1 : N - 1, 1 : N - 1) - (L^o - \text{diag}(\ell)) - \frac{1}{\sigma^2} * \text{eye}(N - 1) \right\|_F^2 \right. \\ & \left. + \|\hat{\theta}(N, 1 : N - 1) - \ell\|_F^2 + \|\hat{\theta}(1 : N - 1, N) - \ell'\|_F^2 + \left(\hat{\theta}(N, N) + \text{sum}(\ell) - \frac{1}{\sigma^2} \right)^2 \right] \\ \text{s.t. } & \begin{cases} -\text{sum}(\ell) = \frac{1}{2} \left(\text{tr}(\hat{\theta}) - \frac{N}{\sigma^2} - \text{tr}(L^o) \right), & \text{(a)} \\ \ell \leq 0, & \text{(b)} \end{cases} \end{aligned}$$

▷ Update L using ℓ :

$$\begin{aligned} L(1 : N - 1, 1 : N - 1) &= L^o - \text{diag}(\ell) \\ L(N, 1 : N - 1) &= \ell \\ L(1 : N - 1, N) &= \ell' \\ L(N, N) &= -\text{sum}(\ell) \end{aligned}$$

▷ Y -step: Learning equivalent representation of the Signals.

$$\min_Y \|X - Y\|_F^2 + \mu_1 \cdot \|Y\|_F^2$$

end while

Trick to Make even More Faster: To Eliminate the second hard constraint $\ell \leq 0$, we will replace the ℓ in the above formulation with $-\rho^2$. Thus in the newer formulation, we would optimize over ρ vector instead of ℓ

Chapter 6

Results on Synthetic Experiments

6.1 Graphs With Sparse Signal Generation Model

DGM										
Author	precision	recall	f_measure	specificity	rho_L_hat_gL	rho_W_hat_gW	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	1.0000	1.0000	1.0000	1.0000	0.9966	0.9821	0.2591	0.1231	0.2140	0.2513
Dong	0.2571	1.0000	0.4091	0.0877	0.7945	0.1513	1.0118	0.4588	0.8355	0.9364
Humbert	0.4219	1.0000	0.5934	0.5673	0.9669	0.8744	0.9339	0.3972	0.7712	0.8109
Siripuram	0.2545	0.5185	0.3415	0.5205	0.7557	0.0645	1.0400	0.4657	0.8588	0.9507
Palomar MCP	0.9000	1.0000	0.9474	0.9649	0.9929	0.9672	7.7172	3.2152	6.3722	6.5631
Palomar Re1	0.8438	1.0000	0.9153	0.9415	0.9890	0.9689	7.0051	2.9315	5.7843	5.9840
Palomar SCAD	0.9000	1.0000	0.9474	0.9649	0.9929	0.9672	7.7172	3.2152	6.3722	6.5631
Sandeep ADMM	0.3103	1.0000	0.4737	0.2982	0.9768	0.9411	6.2388	2.2513	5.1515	4.5954
Ortega CGL R1	0.4355	1.0000	0.6067	0.5906	0.8923	0.7931	2.1294	1.1714	1.7583	2.3912
Ortega CGL R2	0.4355	1.0000	0.6067	0.5906	0.8923	0.7931	2.1294	1.1714	1.7583	2.3912
Nie CLR	0.7105	1.0000	0.8308	0.8713	0.9817	0.8962	0.8668	0.3749	0.7157	0.7652
Sandeep SGL n	1.0000	0.9630	0.9811	1.0000	0.9812	0.9240	5.1127	2.3119	4.2217	4.7191
Sandeep SGL qp	0.9310	1.0000	0.9643	0.9766	0.9898	0.9654	5.5261	2.3478	4.5631	4.7925

SBM										
Author	precision	recall	f_measure	specificity	rho_L_hat_gL	rho_W_hat_gW	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.9783	1.0000	0.9890	0.9935	0.9948	0.9787	0.3469	0.1537	0.2985	0.3241
Dong	0.5570	0.9778	0.7097	0.7742	0.9800	0.8713	0.9142	0.3891	0.7869	0.8202
Humbert	0.6197	0.9778	0.7586	0.8258	0.9673	0.8415	0.9943	0.4204	0.8557	0.8863
Siripuram	0.2481	0.7111	0.3678	0.3742	0.8258	0.0806	0.9256	0.4471	0.7967	0.9426
Palomar MCP	0.7500	1.0000	0.8571	0.9032	0.9878	0.9280	8.9037	3.4744	7.6631	7.3247
Palomar Re1	0.8750	0.9333	0.9032	0.9613	0.9869	0.9255	8.6317	3.6470	7.4290	7.6886
Palomar SCAD	0.7500	1.0000	0.8571	0.9032	0.9878	0.9280	8.9037	3.4744	7.6631	7.3247
Sandeep ADMM	0.3577	0.9778	0.5238	0.4903	0.8958	0.6956	2316.1113	1184.2521	1993.3912	2496.6227
Ortega CGL R1	0.3577	0.9778	0.5238	0.4903	0.9558	0.8407	3.8115	1.7873	3.2804	3.7680
Ortega CGL R2	0.3577	0.9778	0.5238	0.4903	0.9558	0.8407	3.8115	1.7873	3.2804	3.7680
Nie CLR	0.9706	0.7333	0.8354	0.9935	0.8996	0.7040	0.9050	0.3812	0.7789	0.8036
Sandeep SGL n	1.0000	0.8222	0.9024	1.0000	0.9029	0.7096	1.4213	0.9041	1.2232	1.9060
Sandeep SGL qp	1.0000	0.9333	0.9655	1.0000	0.9863	0.9196	7.6712	3.3846	6.6023	7.1354

WS										
Author	precision	recall	f_measure	specificity	rho_L_hat_gL	rho_W_hat_gW	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	1.0000	0.9750	0.9873	1.0000	0.9886	0.9642	0.4725	0.2133	0.4678	0.4770
Dong	1.0000	0.4000	0.5714	1.0000	0.7678	0.4921	0.7806	0.3792	0.7729	0.8479
Humbert	0.9310	0.6750	0.7826	0.9875	0.8694	0.7423	0.9160	0.4072	0.9070	0.9105
Siripuram	0.2022	0.9250	0.3318	0.0875	0.8938	-0.0439	0.7833	0.4271	0.7755	0.9551
Palomar MCP	0.8333	1.0000	0.9091	0.9500	0.9906	0.9485	8.3078	3.5731	8.2260	7.9896
Palomar Re1	0.9286	0.9750	0.9512	0.9813	0.9904	0.9562	8.0811	3.6202	8.0015	8.0949
Palomar SCAD	0.8333	1.0000	0.9091	0.9500	0.9906	0.9485	8.3078	3.5731	8.2260	7.9896
Sandeep ADMM	0.4348	1.0000	0.6061	0.6750	0.9268	0.8053	2848.3548	1411.2474	2820.2915	3155.6452
Ortega CGL R1	0.3448	1.0000	0.5128	0.5250	0.9701	0.9198	5.2134	2.3982	5.1621	5.3626
Ortega CGL R2	0.3448	1.0000	0.5128	0.5250	0.9701	0.9198	5.2134	2.3982	5.1621	5.3626
Nie CLR	0.8696	1.0000	0.9302	0.9625	0.9770	0.9028	0.7603	0.3418	0.7528	0.7643
Sandeep SGL n	1.0000	0.9250	0.9610	1.0000	0.9428	0.8375	2.1581	1.1377	2.1369	2.5439
Sandeep SGL qp	0.8163	1.0000	0.8989	0.9438	0.9911	0.9526	7.4580	3.2173	7.3846	7.1942

Figure 6.1: Standard Graphs: Dorogovtsev Goltsev Mendes(DGM), Stochastic Block Model(SBM), Watts Strogatz(WS), with Sparse Signal Model

BA										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_GL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.9048	1.0000	0.9500	0.9890	0.9782	0.9406	0.3210	0.1726	0.5388	0.5599
Dong	1.0000	0.4737	0.6429	1.0000	0.7193	0.6282	0.4173	0.2405	0.7004	0.7803
Humbert	0.6522	0.7895	0.7143	0.9558	0.8160	0.7452	0.4967	0.2589	0.8336	0.8399
Siripuram	0.2353	0.4211	0.3019	0.8564	0.7831	0.2083	0.5334	0.2997	0.8953	0.9724
Palomar MCP	0.9048	1.0000	0.9500	0.9890	0.9953	0.9880	5.9039	3.0435	9.9088	9.8743
Palomar Re1	0.8636	1.0000	0.9268	0.9834	0.9984	0.9967	4.7061	2.4562	7.8985	7.9690
Palomar SCAD	0.9048	1.0000	0.9500	0.9890	0.9953	0.9880	5.9039	3.0435	9.9088	9.8743
Sandeep ADMM	0.1900	1.0000	0.3193	0.5525	0.8504	0.7687	1939.4565	1182.9735	3255.1115	3838.0728
Ortega CGL R1	0.1712	1.0000	0.2923	0.4917	0.9379	0.9087	3.4654	1.9442	5.8163	6.3078
Ortega CGL R2	0.1583	1.0000	0.2734	0.4420	0.9434	0.9154	3.6554	2.0429	6.1351	6.6279
Nie CLR	0.4524	1.0000	0.6230	0.8729	0.9719	0.9375	0.3225	0.1789	0.5413	0.5803
Sandeep SGL n	0.8947	0.8947	0.8947	0.9890	0.9694	0.9296	4.3928	2.4944	7.3727	8.0931
Sandeep SGL qp	0.9000	0.9474	0.9231	0.9890	0.9474	0.8671	3.7080	2.0922	6.2234	6.7878

Figure 6.2: Standard Graphs: Barabasi Albert (BA), with Sparse Signal Model

ER										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_GL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.9487	0.9737	0.9610	0.9877	0.9844	0.9398	0.4796	0.2182	0.4760	0.5005
Dong	0.4691	1.0000	0.6387	0.7346	0.9658	0.8588	0.7678	0.3491	0.7621	0.8009
Humbert	0.8214	0.6053	0.6970	0.9691	0.8500	0.6537	0.9233	0.4014	0.9164	0.9208
Siripuram	0.1900	1.0000	0.3193	0.0000	0.8515	0.0293	0.8004	0.4148	0.7945	0.9517
Palomar MCP	0.6792	0.9474	0.7912	0.8951	0.9652	0.8483	7.3608	3.0420	7.3062	6.9789
Palomar Re1	0.9667	0.7632	0.8529	0.9938	0.9472	0.8153	5.5187	2.8618	5.4778	6.5655
Palomar SCAD	0.6792	0.9474	0.7912	0.8951	0.9652	0.8483	7.3608	3.0420	7.3062	6.9789
Sandeep ADMM	0.2787	0.8947	0.4250	0.4568	0.7339	0.5619	1204.9842	713.6425	1196.0473	1637.2082
Ortega CGL R1	0.3333	0.9474	0.4932	0.5556	0.9030	0.7989	2.2343	1.2090	2.2177	2.7736
Ortega CGL R2	0.3109	0.9737	0.4713	0.4938	0.9098	0.8071	2.3232	1.2391	2.3060	2.8426
Nie CLR	0.7660	0.9474	0.8471	0.9321	0.9303	0.7179	0.7454	0.3486	0.7398	0.7997
Sandeep SGL n	0.9697	0.8421	0.9014	0.9938	0.9647	0.8624	6.1362	3.0756	6.0907	7.0560
Sandeep SGL qp	0.9677	0.7895	0.8696	0.9938	0.9424	0.8296	4.5933	2.3795	4.5592	5.4588

FF										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_GL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.8974	1.0000	0.9459	0.9758	0.9816	0.9453	0.3703	0.1717	0.3769	0.4104
Dong	0.2536	1.0000	0.4046	0.3758	0.8965	0.6976	0.7759	0.3642	0.7898	0.8707
Humbert	0.7667	0.6571	0.7077	0.9576	0.8276	0.6777	0.8931	0.3807	0.9092	0.9100
Siripuram	0.1750	1.0000	0.2979	0.0000	0.8205	-0.0129	0.7932	0.4003	0.8074	0.9569
Palomar MCP	0.8140	1.0000	0.8974	0.9515	0.9694	0.9094	7.4205	3.2673	7.5539	7.8104
Palomar Re1	0.7895	0.8571	0.8219	0.9515	0.9572	0.8519	7.1021	3.3070	7.2297	7.9052
Palomar SCAD	0.8140	1.0000	0.8974	0.9515	0.9694	0.9094	7.4205	3.2673	7.5539	7.8104
Sandeep ADMM	0.3511	0.9429	0.5116	0.6303	0.7935	0.7077	2096.9167	1121.1437	2134.6050	2680.0460
Ortega CGL R1	0.2846	1.0000	0.4430	0.4667	0.9413	0.8773	4.7891	2.2236	4.8752	5.3154
Ortega CGL R2	0.2810	0.9714	0.4359	0.4727	0.9422	0.8784	4.8307	2.2410	4.9175	5.3569
Nie CLR	0.7174	0.9429	0.8148	0.9212	0.9584	0.8720	0.7273	0.3138	0.7404	0.7500
Sandeep SGL n	0.8974	1.0000	0.9459	0.9758	0.9763	0.9371	5.7655	2.5784	5.8691	6.1636
Sandeep SGL qp	1.0000	0.8286	0.9063	1.0000	0.9595	0.8597	5.6753	2.8477	5.7773	6.8072

SK										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_GL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.8333	0.9167	0.8730	0.9639	0.9548	0.8594	0.5818	0.2508	0.5775	0.6181
Dong	0.4545	1.0000	0.6250	0.7635	0.9567	0.7757	0.8045	0.3449	0.7985	0.8501
Humbert	0.7000	0.7000	0.7000	0.9409	0.8744	0.6603	0.9296	0.3790	0.9227	0.9342
Siripuram	0.1567	0.7833	0.2611	0.1691	0.8500	-0.0479	0.8357	0.3934	0.8295	0.9698
Palomar MCP	0.5566	0.9833	0.7108	0.8456	0.9613	0.7996	8.4460	3.0532	8.3830	7.5253
Palomar Re1	0.8254	0.8667	0.8455	0.9639	0.9620	0.8364	8.3667	3.5836	8.3043	8.8328
Palomar SCAD	0.5566	0.9833	0.7108	0.8456	0.9613	0.7996	8.4460	3.0531	8.3830	7.5253
Sandeep ADMM	0.7000	0.8167	0.7538	0.9310	0.8021	0.6186	2.4940	1.3634	2.4754	3.3604
Ortega CGL R1	0.2611	0.9833	0.4126	0.4516	0.8815	0.7207	4.0852	1.8641	4.0547	4.5945
Ortega CGL R2	0.2611	0.9833	0.4126	0.4516	0.8815	0.7207	4.0852	1.8641	4.0547	4.5945
Nie CLR	0.7761	0.8667	0.8189	0.9507	0.9365	0.7131	0.7876	0.3347	0.7818	0.8250
Sandeep SGL n	0.7931	0.7667	0.7797	0.9606	0.7976	0.6216	2.4847	1.3749	2.4661	3.3888
Sandeep SGL qp	0.7000	0.9333	0.8000	0.9212	0.9601	0.8155	7.9430	3.2353	7.8837	7.9742

Figure 6.3: Standard Graphs: Erdos Renyi(ER), Forest Fire(FF), Stochastic Kronecker(SK), with Sparse Signal Model

6.2 Graphs With Smooth Signal Generation Model

To ensure a comprehensive evaluation, we extended our testing to include graphs with a smooth signal generation model, complementing our previous comparisons that focused on the sparse signal generation model. This additional test provides a broader perspective on the performance of various techniques across different signal types. The results of our extensive comparisons are compelling. Our Proposed Algorithm consistently outperforms all other methods evaluated, both in sparse and smooth signal scenarios. Even in rare instances of close competition, our approach demonstrates a superior ability to learn graphs that more closely align with the ground truths. This consistent performance across varied conditions reinforces the reliability and accuracy of our method in representing the underlying graph structures.

SBM-1										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_gL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	1.0000	0.9778	0.9888	1.0000	0.9925	0.9723	0.4960	0.2086	0.4269	0.4398
Dong	0.6769	0.9778	0.8000	0.8645	0.9879	0.9200	0.9085	0.3814	0.7819	0.8040
Humbert	0.5181	0.9556	0.6719	0.7419	0.9573	0.7472	0.9781	0.4227	0.8418	0.8912
Siripuram	0.2368	1.0000	0.3830	0.0645	0.8865	0.0241	0.9643	0.4527	0.8300	0.9543
Palomar MCP	0.9574	1.0000	0.9783	0.9871	0.9981	0.9888	0.0743	0.0629	0.0640	0.1327
Palomar Rel1	1.0000	0.9556	0.9773	1.0000	0.9935	0.9572	0.1341	0.1231	0.1154	0.2595
Palomar SCAD	0.9574	1.0000	0.9783	0.9871	0.9981	0.9888	0.0743	0.0629	0.0640	0.1327
Sandeep ADMM	0.5769	1.0000	0.7317	0.7871	0.9975	0.9847	0.0823	0.0739	0.0708	0.1557
Ortega CGL R1	0.4054	1.0000	0.5769	0.5742	0.9865	0.9515	0.3999	0.1778	0.3442	0.3748
Ortega CGL R2	0.3947	1.0000	0.5660	0.5548	0.9866	0.9518	0.3978	0.1770	0.3424	0.3732
Nie CLR	0.8235	0.9333	0.8750	0.9419	0.9547	0.8261	0.9030	0.3800	0.7772	0.8011
Sandeep SGL n	0.9783	1.0000	0.9890	0.9935	0.9970	0.9844	0.1193	0.0792	0.1027	0.1669
Sandeep SGL qp	0.9783	1.0000	0.9890	0.9935	0.9972	0.9854	0.0912	0.0715	0.0785	0.1507

WS-1										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_gL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	1.0000	1.0000	1.0000	1.0000	0.9973	0.9904	0.3986	0.1779	0.3947	0.3979
Dong	0.9302	1.0000	0.9639	0.9813	0.9845	0.9150	0.7597	0.3397	0.7522	0.7597
Humbert	1.0000	0.6500	0.7879	1.0000	0.8720	0.7625	0.9126	0.4032	0.9036	0.9016
Siripuram	0.2105	1.0000	0.3478	0.0625	0.9084	0.1012	0.8073	0.4263	0.7993	0.9532
Palomar MCP	0.7273	1.0000	0.8421	0.9063	0.9977	0.9895	0.0691	0.0584	0.0684	0.1306
Palomar Rel1	0.9524	1.0000	0.9756	0.9875	0.9982	0.9934	0.0608	0.0459	0.0602	0.1027
Palomar SCAD	0.7273	1.0000	0.8421	0.9063	0.9977	0.9895	0.0691	0.0584	0.0684	0.1306
Sandeep ADMM	0.5000	1.0000	0.6667	0.7500	0.9977	0.9894	0.0695	0.0589	0.0688	0.1318
Ortega CGL R1	0.4255	1.0000	0.5970	0.6625	0.9877	0.9602	0.2725	0.1390	0.2698	0.3108
Ortega CGL R2	0.3670	1.0000	0.5369	0.5688	0.9921	0.9711	0.1929	0.1088	0.1910	0.2434
Nie CLR	0.8696	1.0000	0.9302	0.9625	0.9847	0.9240	0.7598	0.3405	0.7523	0.7614
Sandeep SGL n	0.9302	1.0000	0.9639	0.9813	0.9940	0.9668	0.1566	0.1083	0.1551	0.2423
Sandeep SGL qp	0.9756	1.0000	0.9877	0.9938	0.9944	0.9698	0.1536	0.1039	0.1521	0.2323

BA-1										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_gL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	1.0000	1.0000	1.0000	1.0000	0.9985	0.9967	0.2762	0.1412	0.4636	0.4582
Dong	1.0000	0.7368	0.8485	1.0000	0.8493	0.7375	0.3541	0.2032	0.5943	0.6593
Humbert	1.0000	0.8947	0.9444	1.0000	0.9635	0.9331	0.4851	0.2509	0.8142	0.8139
Siripuram	0.2143	0.1579	0.1818	0.9392	0.5874	0.1208	0.5735	0.3049	0.9625	0.9892
Palomar MCP	0.9500	1.0000	0.9744	0.9945	0.9994	0.9986	0.0199	0.0155	0.0333	0.0503
Palomar Rel1	1.0000	0.8947	0.9444	1.0000	0.9589	0.9172	0.4689	0.2407	0.7871	0.7811
Palomar SCAD	0.9500	1.0000	0.9744	0.9945	0.9994	0.9986	0.0198	0.0155	0.0333	0.0503
Sandeep ADMM	0.4130	1.0000	0.5846	0.8508	0.9990	0.9976	0.0418	0.0270	0.0701	0.0877
Ortega CGL R1	0.2262	1.0000	0.3689	0.6409	0.9899	0.9788	0.0851	0.0602	0.1429	0.1954
Ortega CGL R2	0.1979	1.0000	0.3304	0.5746	0.9900	0.9789	0.0844	0.0603	0.1416	0.1955
Nie CLR	0.6786	1.0000	0.8085	0.9503	0.9631	0.9203	0.3122	0.1683	0.5240	0.5462
Sandeep SGL n	0.9500	1.0000	0.9744	0.9945	0.9899	0.9820	0.1430	0.0743	0.2400	0.2410
Sandeep SGL qp	0.9756	1.0000	0.9877	0.9938	0.9944	0.9698	0.1536	0.1039	0.1521	0.2323

Figure 6.4: Standard Graphs: SBM, WS, BA with Smooth Signal Model

DGM-1										
Author	precision	recall	f_measure	specificity	rho_L_hat_gL	rho_W_hat_gW	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	1.0000	1.0000	1.0000	1.0000	0.9967	0.9833	0.3809	0.1590	0.3145	0.3245
Dong	0.9310	1.0000	0.9643	0.9766	0.9779	0.9204	0.8924	0.3578	0.7369	0.7304
Humbert	1.0000	0.5185	0.6829	1.0000	0.8086	0.6682	1.0632	0.4277	0.8779	0.8730
Siripuram	0.2571	1.0000	0.4091	0.0877	0.7944	0.1494	1.0118	0.4588	0.8355	0.9365
Palomar MCP	0.6279	1.0000	0.7714	0.8129	0.9965	0.9805	0.1248	0.0902	0.1030	0.1841
Palomar Rel1	0.8182	1.0000	0.9000	0.9298	0.9973	0.9841	0.1034	0.0784	0.0854	0.1601
Palomar SCAD	0.6279	1.0000	0.7714	0.8129	0.9965	0.9805	0.1248	0.0902	0.1030	0.1840
Sandeep ADMM	0.5094	1.0000	0.6750	0.6959	0.9965	0.9807	0.1266	0.0906	0.1045	0.1849
Ortega CGL R1	0.4407	0.9630	0.6047	0.6140	0.9264	0.8360	0.7294	0.2879	0.6023	0.5877
Ortega CGL R2	0.4237	0.9259	0.5814	0.6023	0.9302	0.8427	0.7147	0.2824	0.5901	0.5765
Nie CLR	0.6923	1.0000	0.8182	0.8596	0.9848	0.9184	0.8661	0.3708	0.7152	0.7569
Sandeep SGL n	0.9643	1.0000	0.9818	0.9883	0.9949	0.9708	0.2579	0.1292	0.2130	0.2637
Sandeep SGL qp	0.9643	1.0000	0.9818	0.9883	0.9948	0.9696	0.2575	0.1303	0.2126	0.2659

Figure 6.5: Standard Graphs: DGM with Smooth Signal Model

FF-1										
Author	precision	recall	f_measure	specificity	rho_L_hat_gL	rho_W_hat_gW	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.9722	1.0000	0.9859	0.9939	0.9931	0.9774	0.3826	0.1659	0.3894	0.3967
Dong	0.9714	0.9714	0.9714	0.9939	0.9636	0.8651	0.7027	0.3067	0.7154	0.7332
Humbert	1.0000	0.6286	0.7719	1.0000	0.8607	0.7470	0.8778	0.3699	0.8935	0.8842
Siripuram	0.1967	0.6857	0.3057	0.4061	0.8185	0.0508	0.8020	0.4007	0.8164	0.9578
Palomar MCP	0.9722	1.0000	0.9859	0.9939	0.9993	0.9965	0.0378	0.0316	0.0385	0.0756
Palomar Rel1	0.9722	1.0000	0.9859	0.9939	0.9991	0.9956	0.0456	0.0358	0.0464	0.0857
Palomar SCAD	0.9722	1.0000	0.9859	0.9939	0.9993	0.9965	0.0377	0.0316	0.0384	0.0756
Sandeep ADMM	0.5000	1.0000	0.6667	0.7879	0.9989	0.9952	0.0536	0.0392	0.0546	0.0938
Ortega CGL R1	0.3398	1.0000	0.5072	0.5879	0.9904	0.9720	0.1834	0.0968	0.1867	0.2314
Ortega CGL R2	0.3182	1.0000	0.4828	0.5455	0.9904	0.9721	0.1827	0.0966	0.1860	0.2309
Nie CLR	0.7447	1.0000	0.8537	0.9273	0.9580	0.8844	0.7227	0.3122	0.7357	0.7462
Sandeep SGL n	0.9722	1.0000	0.9859	0.9939	0.9961	0.9873	0.1154	0.0660	0.1174	0.1577
Sandeep SGL qp	0.9722	1.0000	0.9859	0.9939	0.9991	0.9958	0.0513	0.0366	0.0522	0.0874

SK-1										
Author	precision	recall	f_measure	specificity	rho_L_hat_gL	rho_W_hat_gW	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.9667	0.9667	0.9667	0.9934	0.9903	0.9563	0.5461	0.2263	0.5420	0.5577
Dong	0.5960	0.9833	0.7421	0.8686	0.9821	0.9019	0.7876	0.3285	0.7818	0.8096
Humbert	0.9322	0.9167	0.9244	0.9869	0.9742	0.9126	0.9029	0.3641	0.8961	0.8974
Siripuram	0.2054	0.3833	0.2674	0.7077	0.8752	0.1121	0.9298	0.3977	0.9229	0.9803
Palomar MCP	0.9231	1.0000	0.9600	0.9836	0.9974	0.9845	0.0731	0.0651	0.0725	0.1604
Palomar Rel1	0.8696	1.0000	0.9302	0.9704	0.9972	0.9833	0.0761	0.0678	0.0755	0.1672
Palomar SCAD	0.9231	1.0000	0.9600	0.9836	0.9974	0.9845	0.0731	0.0651	0.0725	0.1604
Sandeep ADMM	0.4225	1.0000	0.5941	0.7307	0.9956	0.9729	0.0989	0.0914	0.0982	0.2252
Ortega CGL R1	0.3155	0.9833	0.4777	0.5796	0.9732	0.9320	0.3814	0.1647	0.3786	0.4060
Ortega CGL R2	0.3089	0.9833	0.4701	0.5665	0.9735	0.9324	0.3795	0.1641	0.3766	0.4044
Nie CLR	0.9400	0.7833	0.8545	0.9901	0.9210	0.7767	0.7764	0.3248	0.7706	0.8005
Sandeep SGL n	0.9375	1.0000	0.9677	0.9869	0.9947	0.9712	0.1464	0.0962	0.1453	0.2371
Sandeep SGL qp	1.0000	0.8500	0.9189	1.0000	0.9689	0.8754	0.4221	0.1991	0.4190	0.4907

Figure 6.6: Standard Graphs: FF and SK with Smooth Signal Model

ER-1										
Author	precision	recall	f_measure	specificity	$\rho_{L_hat_gL}$	$\rho_{W_hat_gW}$	rmse_L	rmse_W	rel_rmse_L	rel_rmse_W
Our	0.9500	1.0000	0.9744	0.9877	0.9902	0.9706	0.4106	0.1913	0.4075	0.4389
Dong	0.7600	1.0000	0.8636	0.9259	0.9810	0.9143	0.7514	0.3327	0.7458	0.7633
Humbert	0.8444	1.0000	0.9157	0.9568	0.9881	0.9558	0.8528	0.3726	0.8465	0.8548
Siripuram	0.1868	0.8947	0.3091	0.0864	0.8574	-0.0472	0.8248	0.4181	0.8187	0.9591
Palomar MCP	0.6909	1.0000	0.8172	0.8951	0.9978	0.9903	0.0665	0.0549	0.0660	0.1259
Palomar Rel1	0.9744	1.0000	0.9870	0.9938	0.9983	0.9932	0.0595	0.0461	0.0590	0.1058
Palomar SCAD	0.9744	1.0000	0.9870	0.9938	0.9983	0.9929	0.0616	0.0476	0.0611	0.1092
Sandeep ADMM	0.4043	1.0000	0.5758	0.6543	0.9977	0.9898	0.0687	0.0568	0.0682	0.1304
Ortega CGL R1	0.3895	0.9737	0.5564	0.6420	0.9804	0.9440	0.3654	0.1666	0.3627	0.3823
Ortega CGL R2	0.3627	0.9737	0.5286	0.5988	0.9817	0.9467	0.3515	0.1614	0.3489	0.3703
Nie CLR	0.8919	0.8684	0.8800	0.9753	0.9098	0.7111	0.7380	0.3470	0.7325	0.7960
Sandeep SGL n	0.9744	1.0000	0.9870	0.9938	0.9950	0.9769	0.1554	0.0956	0.1542	0.2194
Sandeep SGL qp	0.9744	1.0000	0.9870	0.9938	0.9962	0.9867	0.0877	0.0639	0.0870	0.1466

Figure 6.7: Standard Graphs: ER with Smooth Signal Model

Chapter 7

Conclusion & Future Work

In this thesis, we introduced a novel Graph Learning Algorithm designed to infer the graphical relationships underlying data defined at each node. Addressing the inconsistencies observed in other models regarding hyperparameter tuning, we developed a principled approach to perform hyperparameter tuning. This method is not only compatible with our model but also extends its applicability to other proposed models, enhancing the robustness and adaptability of the hyperparameter tuning process across various graph learning frameworks.

Recognizing the complexities in model convergence, we proposed a novel streaming algorithm aimed at efficiently modifying the graph when new nodes are added to an existing, already-learned graph. This innovation ensures the scalability and dynamism of the graph learning process. We specifically applied this streaming algorithm to Dong et al.’s model and rigorously tested its performance, demonstrating its effectiveness in maintaining the integrity and accuracy of the graph structure as it evolves.

Furthermore, we proposed a streaming variant of our own graph learning algorithm, although this work is still in progress. Our future aspirations include developing a robust streaming algorithm that seamlessly integrates with our proposed model, ensuring it can handle the dynamic nature of real-world data. We envision this advancement will significantly contribute to the field of graph learning, providing a scalable and efficient solution for continuously evolving data environments.

Chapter 8

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